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COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)

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COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)  
05/25

PART 1 GENERAL

1.1 DEFINITIONS

The cost-loaded Network Analysis Schedule (NAS) is a tool to manage the project, both for Contractor and Government activities. The NAS is also used to report progress, evaluate time extensions, and provide the basis for progress payments.

For consistency, when scheduling software terminology is used in this section, the terms in Oracle Primavera's scheduling programs are used.

1.2 NAS REQUIREMENTS PRIOR TO THE START OF WORK

1.2.1 Preliminary Scheduling Meeting

Before preparation of the Baseline NAS, and prior to the start of work, meet with the Contracting Officer to discuss the proposed NAS and the requirements of this section. Propose projected data dates for monthly update NAS for the project and incorporate each monthly update submittal into the submittal register. Discuss required forms, terminology, and submittal requirements of this section and other requirements related to schedule management for this contract. Use the NAVFAC-provided Preliminary Scheduling Meeting Guideline in the completion of the intended mutual understanding.

1.2.2 Baseline NAS

Submit the Baseline NAS within 30 calendar days after contract award. Data date must be set to contract award date and no progress recorded for any activity. Only bonds and the required Preliminary Scheduling Meeting may be recorded as complete prior to acceptance of the Baseline NAS. The acceptance of a Baseline NAS is a condition precedent to:

- a. The Contractor starting demolition work or construction stage(s) of the contract.
- b. Processing Contractor's invoices(s) for any items other than bonds.
- c. Review of any NAS updates. The contractor is still required to submit monthly NAS updates. Without an accepted baseline NAS, the Government may return Monthly NAS update as received for record.

Submittal of the Baseline NAS is the Contractor's certification that the submitted NAS meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with baseline NAS submittal.

### 1.3 WEEKLY LOOK AHEAD SCHEDULE

#### 1.3.1 Quality Control and Production Integration with NAS

Submit the first Look Ahead Schedule 30 days after contract award and continue to submit weekly until contract completion.

Transmit the Weekly Look Ahead Schedule Submittal to the Contracting Officer on the first day of the current week that the Look Ahead Schedule covers and at least one working day prior to any Quality Control meetings. Contractor is required to provide all attendees at meetings with a copy of the Look Ahead Schedule. A Look Ahead Schedule is required to be submitted weekly regardless of whether a meeting is held or not.

#### 1.3.2 Weekly Look Ahead Schedule Requirements

Prepare and issue a Three-Week Look Ahead Schedule to provide a more detailed day-to-day plan of upcoming work identified on the current Network Analysis Schedule. Requirements include:

- a. For each Look Ahead Schedule activity, identify parent NAS activity number(s). The parent NAS activity is the activity in the NAS that would incorporate the Look Ahead Schedule activity requirement and scope of work.
- b. Update schedule each week to show the planned work for the following three-week period. Also include previous week, as-built work, showing actual start and finish dates.
- c. Include upcoming outages, closures, preparatory meetings, and initial meetings, testing and inspections.
- d. Identify any offsite fabrication and production activities.
- e. Clearly identify Longest Path activities on the Weekly Look Ahead Schedule. Include a key or legend that distinguishes longest path activities. Include all Longest Path activity NAS start/finish dates exceeded and occurring during this period.
- f. Identify responsibility for each activity.
- g. The detail work plans are to be bar chart type schedules, derived from but maintained separately from the Project NAS on an electronic spreadsheet program and printed on 299 by 432 mm sheets as directed by the Contracting Officer.
- h. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work.
- i. Look Ahead activities are not limited to only onsite work. Include submittal, offsite fabrication, and procurement activities. Include design activities, meetings, and coordination when part of scope.
- j. Upload the Weekly Look Ahead Schedules to eCMS as a submittal by the date and time required.

#### 1.4 MONTHLY NETWORK ANALYSIS

Submittal of Monthly NAS is the Contractor's certification that the submitted NAS meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with the monthly NAS submittal.

##### 1.4.1 Monthly Update Meeting

Meet with Government representative(s) at monthly intervals to review and agree on the information presented in the Monthly NAS Update. The Monthly Update Meeting must be a separate and distinct meeting from other meetings required of the contract. As a minimum, accomplish the following with each meeting held monthly:

- a. Prior to the meeting, provide an agenda of proposed topics to be discussed to include but not limited to narrative report and items listed in paragraph REQUIRED TABULAR REPORTS. Document agenda and minutes with Meeting Minutes object in eCMS for each monthly update meeting.
- b. The Approved Designated Project Scheduler must present the NAS data using Oracle Primavera P6 software and have baseline NAS, monthly update NAS, and Time Impact Analysis native files available to review as necessary.
- c. Review Time Impact Analysis.
- d. Document action items in eCMS Meeting Minutes for necessary revisions to Monthly NAS Update and to address any delays and or problem areas in a timely manner.

##### 1.4.2 Monthly NAS Updates

- a. The submission of an accepted, updated NAS to the Government is a condition precedent to the processing of the Contractor's invoice.
- b. Regardless of whether an invoice is being submitted monthly, or no project progress occurred, an updated NAS (or with new data date) must be submitted monthly to the Government. The Monthly NAS update must be submitted within 5 calendar days of the data date.
- c. Regardless of whether a baseline or other NAS is submitted and accepted, an updated NAS must be submitted monthly to the Government. Monthly updates submitted before the baseline being accepted may be returned as received for record.
- d. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES, with each monthly NAS update submittal.
- e. Activity progress must incorporate as-built events as they occurred and correspond to records including but not limited to submittals and daily production and quality control reports.
- f. Software Settings: Handle NAS calculations through Retained Logic, not Progress Override. Out of Sequence progress is not allowed. Relationships must be changed to reflect as-built relationships. Show all activity durations and float values in days. Show activity

progress using Remaining Duration. Set default activity type to "Task Dependent".

- g. Updated NAS must reflect current Contract Completion Date and contract value in accordance with all conformed contract modifications issued prior to data date of NAS update.

#### 1.4.3 As-Built NAS

As a condition precedent to the release of retention and making final payment, submit an "As-Built NAS," as the last NAS update showing all activities at 100 percent completion. This NAS must reflect the exact manner in which the project was actually constructed.

#### 1.5 CORRESPONDENCE AND TEST REPORTS

Reference NAS Activity IDs that are being addressed in each correspondence (e.g., letters, Submittals, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) and test report (e.g., concrete, soil compaction, weld, pressure).

#### 1.6 ADDITIONAL NETWORK ANALYSIS SCHEDULE REQUIREMENTS

Other specification sections may include additional NAS related requirements, including systems to be inspected, tested and commissioned, and submittal procedures. Those requirements must be incorporated into the NAS.

#### 1.7 SUBMITTALS

Government approval/acceptance is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

Baseline NAS; G

Designated Project Scheduler; G

##### SD-07 Certificates

Weekly Look Ahead Schedule

Monthly NAS Updates; G

##### SD-11 Closeout Submittals

As-Built NAS; G

#### 1.8 SOFTWARE

Prepare and maintain project NAS using Oracle Primavera P6 software in a version compatible with Government's current version. Submit XER files exported as version 17 or earlier. Importing data into P6 using data conversion techniques or third party software is cause for rejection of the submitted NAS. NAS with Performing Organizational Breakdown Structure

(POBS) data in any form other than column is cause for rejection. NAS must be scheduled by using the Schedule command after last entry in Oracle Primavera P6 prior to export for submission to the Government.

## 1.9 DESIGNATED PROJECT SCHEDULER

Within 30 calendar days of contract award, submit to the Contracting Officer for approval an individual who will serve as the Designated Project Scheduler. Include a copy of the candidate's resume with qualifications. The Designated Project Scheduler is subject to removal by the Contracting Officer for non-compliance with the requirements specified in the Contract. Payment request will not be processed without an approved Designated Project Scheduler.

### 1.9.1 Qualifications

The Designated Project Scheduler must have served as a scheduler and prepared and maintained at least three previous NAS for projects, at least one within the last 5 years of similar size and complexity to this contract, using Oracle Primavera P6.

### 1.9.2 Duties

Duties of the Designated Project Scheduler include to be actively involved and be responsible for the following:

- a. Prepare Baseline NAS.
- b. Prepare monthly NAS updates.
- c. Prepare tabular reports.
- d. Prepare Time Impact Analysis (TIA) as necessary.
- e. Provide certification of contract compliance for each NAS and TIA submittal transmitted to government for review.
- f. Participate with the Prime Contractor and Government Representative in a monthly teleconference call, and scheduled with sufficient time to support the Monthly Network Analysis Updates process, to discuss project status, NAS updates, critical activities, potential delays, and contract modifications impacting any project progress. Have a computer with P6 software available during the meeting.

## 1.10 NETWORK SYSTEM FORMAT

Prepare the NAS in accordance with the following Primavera P6 settings and parameters. Deviation from these settings and parameters, without prior consent of the Contracting Officer, is cause for rejection of NAS submittal.

### 1.10.1 NAS Activity Properties and Level of Detail

#### 1.10.1.1 Activity Identification and Organization

- a. Identify construction activities planned for the project and other activities that could impact project completion if delayed in the NAS.
- b. Each activity must have a unique name.

- c. Identify administrative type activity/milestones, including all pre-construction submittal and permit requirements prior to demolition or construction stage.
- d. Include durations for procurement, Contractor quality control, acceptance testing and training in the NAS.
- e. Include the Government approval time required for the submittals that require Government Approval prior to construction, as indicated in [Section 01 33 00 SUBMITTAL PROCEDURES](#).
- f. Create separate activities for each Phase, Area, Floor Level and Location the activity is occurring.
- g. Do not use activities to represent non-work type reference (e.g., Serial Letter, Request for Information) in NAS. When RFIs need to be included in NAS, identify Non-work reference with use of added Activity codes (e.g, RFI code, Serial Letter Code). Document use of added activity codes in narrative report.

Activity categories included in the NAS are specified below.

#### 1.10.1.2 Activity Logic

- a. With the exception of the Contract Award and Contract Completion Date (CCD) milestone activities, activity must not be open-ended; each activity must have at least one predecessor and at least one successor.
- b. Activities must not have open start or open finish (dangling) logic.
- c. Do not use lead or lag logic without Contracting Officer prior approval. Request for approval must be submitted at the required Preliminary Scheduling Meeting for use in the Baseline NAS and at the required Monthly Network Analysis Update Meeting for use in a monthly updated NAS.
- d. Minimize redundant logic ties.
- e. Once an activity exists on the NAS it must not be deleted or renamed to change the scope of the activity and must not be removed from the NAS logic without approval from the Contracting Officer. Request for approval must be submitted with the required Monthly NAS Update Narrative.
- f. While an activity cannot be deleted, where said activity is no longer applicable to the NAS, but must remain within the logic stream for historical record, change the activity original and remaining duration to zero and clearly label "(NO LONGER REQUIRED)" after the Activity Name.
  - (1) Document any such change in the activities' "Notebook," including a date and explanation for the change.
  - (2) The ID number for a "NO LONGER REQUIRED" activity must not be re-used for another activity.
- g. No activity can be added to the NAS without the prior approval from the Contracting Officer. Request for approval must be submitted at

the required Monthly Network Analysis Update Narrative for use in a monthly updated NAS.

- h. Activity Type for all activities except for milestones must be set to "Task Dependent". "Level of Effort" Activity Type is not allowed.
- i. Duration Type for all activities except for milestones must be set to "Fixed Duration & Units".
- j. Percent Complete Type for all activities including milestones must be set to "Physical".

#### 1.10.1.3 Longest Path Activity Baseline Limitation

For P6 settings, critical activities are defined as being on the Longest Path. Longest Path (Critical) Activities must not make up more than 30 percent of all activity within the Construction Baseline NAS.

#### 1.10.1.4 Assigned Calendars

All NAS activity must be assigned calendars that reflect required and anticipated non-work days.

Total work hours/day for all defined calendars is set to 8.

#### 1.10.1.5 Activity Categories

##### 1.10.1.5.1 Pre-construction Activities

Examples of pre-construction activities include, but are not limited to, bond approval, permits, pre-construction submittals and approvals. Include pre-construction activities that are required to be completed prior to the Contractor starting the demolition or construction stage of work.

##### 1.10.1.5.2 Procurement Activities

Include activities necessary to manage the ordering, fabrication, and delivery of material and equipment. Procurement activities must not have a duration in excess of 60 calendar days without prior approval by the Contracting Officer. Include within sequence of off-site assembly and fabrication any quality control and testing being performed.

Examples of procurement activities include, but are not limited to: Material/equipment submittal preparation, submittal and approval of material/equipment; material/equipment fabrication and delivery, and material/equipment on-site. As a minimum, separate procurement activities must be provided for critical items, long lead items, items requiring Government approval and material/equipment procurement for which payment will be requested in advance of installation. Show each delivery with relationship tie to the Construction Activity specifically for the delivery.

##### 1.10.1.5.3 Government Activities

Government and other agency activities that could impact progress must be clearly identified. Government activities include, but are not limited to; Government approved submittal reviews, Government conducted inspections/tests, environmental permit approvals by State regulators,

utility outages, and delivery of Government Furnished Material/Equipment.

#### 1.10.1.5.4 Construction Quality Management (CQM) Activities

The Preparatory and Initial Phase meetings for each Definable Feature of Work identified in the Contractor's Quality Control Plan must be included in the Weekly Look Ahead Schedules. Preparatory and Initial phase meetings are not required in the NAS, but can be represented by a start milestone linked to successor parent Construction Activity. The Follow-up Phase must be represented by the Construction Activities themselves in the NAS.

#### 1.10.1.5.5 Construction Activities

On-site construction activities that are the responsibility of the Contractor or Contractor's subcontractors must not have a duration in excess of 20 working days without prior approval by the Contracting Officer. Contractor activities must be driven by calendars that reflect Saturdays, Sundays and all Federal Holidays as non-work days, unless otherwise defined in this contract. Federal Holidays are as defined in Federal Law 5 USC 6103.

#### 1.10.1.5.6 Turnover and Closeout Activities

Include activities or milestones for items on the NAVFAC Red Zone Checklist/POAM that are applicable to this project. As a minimum, include required Contractor testing, required Government acceptance inspections on equipment, Pre-Final Inspection, Punch List Completion, Final Inspection and Acceptance. Add an unconstrained start milestone for the initial NAVFAC Red Zone - Facility Turnover Planning Meeting at approximately 75 percent construction contract completion or six months prior to Contract Completion Date (CCD), whichever is sooner. Include a separate NAVFAC Red Zone - Facility Turnover Planning Meeting Milestone for each phase if the contract requires construction to be completed in phases.

#### 1.10.1.6 Contract Milestones and Constraints

##### 1.10.1.6.1 Project Start Date Milestones

Include as the first activity on the NAS a start milestone titled, "Contract Award", which must have a Mandatory Start constraint equal to the Contract Award Date.

##### 1.10.1.6.1.1 Post-Award Kickoff (PAK) meeting Milestone

Include an unconstrained finish milestone on the NAS titled, "Post-Award Kickoff Meeting". The Post Award Kickoff Meeting may be a single day, or it may range over several days. The intent is to cover all PAK topics, including Partnering and Concept Design Workshop (if required) in one continuous session.

##### 1.10.1.6.2 Pre-Construction Meeting Milestone

Include an unconstrained finish milestone on the NAS titled, "Pre-Construction Meeting". The Pre-Construction meeting may be a single day, or it may range over several days. The intent is to cover all the Pre-Con topics, including Partnering and DD1354.

1.10.1.6.3 Preconstruction Submittals Finish Milestone

Include an unconstrained finish milestone on the NAS titled, "Preconstruction Submittals". This milestone is complete when all required preconstruction submittals have been reviewed and approved by the Government.

1.10.1.6.4 Contractor Mobilization Finish Milestone

Include an unconstrained finish milestone on the NAS titled, "Contractor Mobilization".

1.10.1.6.5 NAVFAC Red Zone - Facility Turnover Planning Meeting Milestones

See paragraph TURNOVER AND CLOSEOUT ACTIVITIES above.

1.10.1.6.6 Substantial Completion Milestone

Include an unconstrained finish milestone on the NAS titled "Substantial Completion." Substantial Completion is defined as the point in time the Government would consider the project ready for beneficial occupancy wherein by mutual agreement of the Government and Contractor, Government use of the facility is allowed while construction access continues in order to complete remaining items (e.g., punch list and other close out submittals). Include a separate Substantial Completion Milestone for each phase if the contract requires construction to be completed in phases.

1.10.1.6.7 DD-1354 Finish Milestone

Add unconstrained finish milestone, titled "DD-1354" and scheduled 30 calendar days prior to Substantial Completion, whenever a Form DD-1354 is required in accordance with Section 01 20 00 PRICE AND PAYMENT PROCEDURES. Include a separate DD-1354 Finish Milestone for each phase if the contract requires construction to be completed in phases.

1.10.1.6.8 Project Completion Milestone

Include an unconstrained finish milestone on the NAS titled "Project Completion." Project Completion is defined as the point in time all contract requirements are complete and verified by the Government with a successful Final Inspection in accordance with Section 01 45 00 QUALITY CONTROL. This milestone must have the Contract Completion Date (CCD) milestone as its only successor.

1.10.1.6.9 Contract Completion Date (CCD) Milestone

Last NAS entry must be an unconstrained finish milestone titled "Contract Completion (CCD: DD-MM-YY)." DD-MM-YYYY is the current contract completion date at data date, day-month-year corresponding to P6 Must Finish By Date. NAS milestone updates of Project Completion finish date for longest path must reflect calculated float as positive or negative based on CCD. Calculation of NAS updates must be such that if the finish of the "Project Completion" milestone falls after the contract completion date, then negative float is calculated on the longest path. If the finish of the "Project Completion" milestone falls before the contract completion date, the float calculation must reflect positive float on the longest path.

#### 1.10.1.6.10 Additional Milestones

Provide up to five additional milestones as required by Contracting Officer.

#### 1.10.1.7 Work Breakdown Structure & Activity Code

At a minimum, establish a Work Breakdown Structure (WBS) and provide activity codes identified as follows:

##### 1.10.1.7.1 Work Breakdown Structure (WBS)

Group all activities and milestones within appropriate WBS elements/levels including, at a minimum, the following:

a. Project Milestones:

- (1) Management Milestones
- (2) Project Administrative Meetings
- (3) Permits

b. Pre-Construction Phase:

- (1) Submittals and Reviews
- (2) Procurement
- (3) Mobilization

c. Construction Phase: Create multiple elements/levels in accordance with project specific definable features of work including in WBS descending order as follows:

- (1) General Area
  - (a) Type of Work Item
    1. Location

d. Project Closeout: Include activity items such as, but not limited to, Punchlist, Demobilization, O&M, As-built Drawings, Training, and As-built NAS.

e. Modifications: Create Conformed and Non-Conformed elements/levels under Modification WBS. Create multiple elements/levels as the project progresses identified by issue and Fragnet placed in Conformed for modifications issued prior data date, or Non-Conformed for issues not modified to contract prior data date.

f. Removed Activity: Activity is "removed" by remaining within logic sequence, eliminating duration and adding "(NO LONGER REQUIRED)" after Activity Name in Activity Table.

g. Commissioning & Testing:

- (1) Specific area/locations of commissioning

(2) Final Testing

(3) Training

1.10.1.7.2 Responsibility Code

Use "RESP" for Activity Code Name. All activities in the project NAS must be identified with the resource, including business name, for completing the task. Activities must not belong to more than one responsible party.

1.10.1.7.3 Activity Category Code

Use "CAT" for Activity Code Name for the following Project Level activity codes:

- a. Assign "PROC" value to Procurement type activity
- b. Assign "PRE-CON" value to Pre-construction activity
- c. Assign "CONS" value to Construction type activity
- d. Assign "TEST" value to dedicated testing type activities
- e. Assign "CX" value to dedicated Commissioning type activities
- f. Assign "CLOS" value to dedicated Close Out type activity
- g. Assign "OTHR" to other activity not otherwise designated

1.10.1.7.4 Additional Activity Codes

Provide up to an additional five activity codes as required by the Contracting Officer

1.10.1.7.5 Drawing Code

Identify all activities in the project NAS with its respective Drawing Code. The Drawing Code is the Sheet Number on the primary project drawing which indicates work to be performed. If an activity does not have an applicable Drawing Code (e.g., Mobilize), the code must be "0000".

1.10.1.7.6 Construction Specification Institute (CSI) Masterformat Code

Identify all activities in the project NAS with its respective Specification Section number. Activities must not belong to more than one Section number. If an activity does not have an applicable CSI Code (e.g., Mobilize), the code must be "0000".

1.10.1.8 Adverse Weather Lost Work Days

Use the National Oceanic and Atmospheric Administration's (NOAA) Summary of Monthly Normals report to obtain the historical average number of days each month with precipitation, using a nominal 30-year, greater than 2.5 mm precipitation amount parameter, as indicated on the Station Report for the NOAA location closest to the project site as the basis for establishing a "Weather Calendar" showing the number of anticipated non-workdays for each month due to adverse weather, in addition to Saturdays, Sundays and all Federal Holidays as non-work days.

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Use the following schedule of anticipated monthly non-work days due to adverse weather as the basis for establishing a "Weather Calendar" showing the number of anticipated non-workdays for each month due to adverse weather, in addition to Saturdays, Sundays and all Federal Holidays as non-work days.

| MONTHLY ANTICIPATED ADVERSE WEATHER LOST WORK DAY |     |     |     |     |     |     |     |     |     |     |     |
|---------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| JAN                                               | FEB | MAR | APR | MAY | JUN | JUL | AUG | SEP | OCT | NOV | DEC |
| 4                                                 | 5   | 5   | 6   | 3   | 1   | 0   | 0   | 1   | 6   | 4   | 5   |

Assign the Weather Calendar to any activity that could be impacted by adverse weather. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time only extension for the difference of days between the anticipated and actual adverse weather delay if the number of actual adverse weather delay days exceeds the number of days anticipated for the month in which the delay occurs and the adverse weather delayed activities are on the longest path to contract completion in the period when delay occurred. A lost workday due to weather conditions is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity. Impacts resulting from adverse weather must be documented in Narrative Report for the month that it occurred.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

#### 1.10.1.9 Anticipated Restricted Delays

Use anticipated non-working days as defined in Section 01 14 00 WORK RESTRICTIONS, use Anticipated Restricted Delays as basis for establishing a "Security Calendar" showing the number of anticipated non-workdays for each month due to anticipated restrictions, in addition to anticipated adverse weather, Saturdays, Sundays and all Federal Holidays as non-work days. Assign the Security Calendar to any activity that could be impacted by restriction delays. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time extension for the difference of days between the anticipated and actual lost work days if the number of actual restriction delay days exceeds the number of anticipated for the month in which the delay occurs and the restriction delayed activities are critical to contract completion. A lost workday due to restriction delay is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity.

Impacts resulting from restriction delays must be documented in Narrative Report for the month that it occurred.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

#### 1.10.1.10 Cost Loading

The Project Network Analysis Schedule (NAS) must be cost-loaded and will provide the basis for progress payments. Earned Value Reports must be

derived from and correspond to cost loaded NAS. Use the Critical Path Method (CPM) and the Precedence Diagram Method (PDM) to satisfy time and cost applications.

Sum of all costs assigned to activities must equal the current contract value as of the data date of the baseline or update NAS. Activity cost loading must be without front-end loading. Provide additional documentation if requested by the Contracting Officer.

#### 1.10.1.10.1 Cost Loading Activities

- a. Cost load construction activities. Define construction activities as on-site work activities that can be traced directly to the construction, alteration, demolition, repair of a project facility, infrastructure, or sitework.
- b. Assign costs for materials which payment will be requested in advance of installation, including their quantities, to their respective procurement activities that were approved by the Contracting Officer. Assign labor costs, including their quantities, for material and equipment paid for after installation to their respective construction activities.
- c. Costs for mobilization will not be paid as individual pay items with the exception of batch plant set-up, mobilization of dredging equipment or other similar labor-intensive situations. If payment for mobilization costs are approved, assign costs for demobilization activities. The value of demobilization activities must not be less than 40 percent of the total costs for mobilization activities. If not approved as a separate line item, evenly disperse over construction activities the costs for mobilization and demobilization.
- d. The value of commissioning, testing, and closeout WBS section may not be less than 10 percent of the total costs for procurement and construction activities.
- e. ALL activities assigned Government responsibility will have Zero Cost. No contractor cost should be assigned to an activity designated as a Government responsibility.
- f. Do not include field overhead positions as individual cost loaded activities. Evenly disperse non-construction costs (e.g., meetings, presentations, submittals, surveys, haul road or storage yard maintenance, dust control, security) and profit to construction activities assigned to the Construction Phase WBS level.
- g. If assigning resources to activities, setting for Calculate costs from units in Resources tab must be unchecked.

#### 1.10.1.10.2 Partial Payment

Breakdown unit of measure and cost must be defined within P6 Activity Detail Expenses for partial payment of any cost loaded activity. Lump sum cost loaded activity will not be partially paid.

#### 1.10.2 NAS Software Settings and Restrictions

- a. Activity Constraints: Date/time constraint(s), other than those required by the contract, are not allowed unless accepted by the

Contracting Officer. Identify any constraints proposed and provide an explanation for the purpose of the constraint in the Narrative Report as described in paragraph REQUIRED TABULAR REPORTS.

- b. Default Progress Data Disallowed: Actual Start is date work begins on activity with intent to pursue work to substantial completion. Actual Finish is date work is substantially complete to point where successor activity can begin. Actual dates on the NAS must correspond with activity dates reported on the Contractor Quality Control and Production Reports.
- c. At a minimum, include the following settings and parameters in P6 NAS preparation:
  - (1) General: Define or establish Calendars and Activity Codes at the "Project" level, not the "Global" level.
  - (2) Admin Drop-Down Menu, Admin Preferences, Time Periods Tab:
    - (a) Set time periods for P6 to 8.0 Hours/Day, 40.0 Hours/Week, 172.0 Hours/Month and 2000.0 Hours/Year.
    - (b) Use assigned calendar to specify the number of work hours for each time period: Must be checked.
  - (3) Admin Drop-Down Menu, Admin Preferences, Earned Value Tab:
    - (a) Technique for computing performance percent complete: Use "Activity percent Complete"
    - (b) Technique for computing Estimate to Complete: Use "PF = 1".
    - (c) Earned Value Calculation: Use "Budgeted values with current dates".
  - (4) Project Level, Dates Tab:
    - (a) Set "Must Finish By" date to "Contract Completion Date", and set "Must Finish By" time to 05:00pm.
  - (5) Project Level, Defaults Tab:
    - (a) Duration Type: Set to "Fixed Duration & Units".
    - (b) Percent Complete Type: Set to "Physical".
    - (c) Activity Type: Set to "Task Dependent".
    - (d) Calendar: Set to "Standard 5 Day Workweek". Calendar must reflect Saturday, Sunday and all Federal holidays as non-work days. Alternative calendars may be used with Contracting Officer approval.
  - (6) Project Level, Calculations Tab:
    - (a) Default Price/Unit for activities without resource or role Price/Units: Set to "\$1/h".
    - (b) Activity percent complete based on activity steps: Must be

Checked.

- (c) Link Budget and At Completion for not started activities:  
Must be Checked.
  - (d) Reset Remaining Duration and Units to Original: Must be  
Selected.
  - (e) Subtract Actual from At Completion: Must be Selected.
  - (f) Recalculate Actual units and Cost when duration percent  
complete changes: Must be Checked.
  - (g) Update units when costs change on resource assignments:  
Must be Unchecked.
  - (h) Link Actual to Date and Actual This Period Units and Cost:  
Must be Checked.
- (7) Project Level, Settings Tab:
- (a) Define Critical Activities: Check "Longest Path".
- (8) Work Breakdown Structure Level, Earned Value Tab:
- (a) Technique for Computing Performance Percent Complete:  
"Activity percent complete" is selected.
  - (b) Technique for Computing Estimate to Complete (ETC): "PF =  
1" is selected.

#### 1.10.3 Required Tabular Reports and Native P6 XER Files

Include the following reports with the Baseline, Monthly Update and any other required NAS submittals:

a. Time Scaled Logic NAS

Provide formatted 11 by 17-inch Time-scaled Logic NAS in color and landscape-oriented with each NAS submittal. Clearly show activities on the longest path setting Gantt chart longest path activity bars to red. Group activities by WBS and sort by finish date in ascending order. Include the following information in column form for each activity and include accompanying Gantt chart:

- (1) Activity ID
- (2) Activity Name
- (3) Original Duration
- (4) Remaining duration
- (5) Physical Percent Complete
- (6) Start Date
- (7) Finish Date

(8) Total Float

(9) Project-level Calendar

- b. Previous Monthly Update Comparison Time Scaled Logic NAS (Submit with all Monthly NAS Update Submittals.)

Provide formatted 11 by 17-inch Time-scaled Logic NAS in color and landscape-oriented with each monthly NAS update submittal. Clearly show activities on the current month longest path setting Gantt chart longest path activities bars to red. Show previous month activities as yellow bars and previous month milestones in yellow within Gantt chart. Sort by finish date in ascending order. Filter activities for longest path. Maintain and assign the accepted previous monthly NAS update or the accepted baseline NAS for the first update submittal as the baseline and primary baseline in P6 before printing the NAS. Include the following information in column form for each activity and include accompanying Gantt chart:

(1) Activity ID

(2) Activity Name

(3) Original Duration

(4) Current Month Remaining Duration

(5) Current Month Start Date

(6) Previous Month Update Start Date (BL Project Start)

(7) Start Date Delta between Current Month and Previous Month  
(Variance - BL Project Start Date)

(8) Current Month Finish Date

(9) Previous Month Finish Date (BL Project Finish)

(10) Finish Date Delta between Current Month and Previous Month  
(Variance - BL Project Start Date)

(11) Current Month Total Float

- c. P6 native XER file: Include the back-up native .xer program file compatible with the current Government version of Primavera P6. Each native NAS file must have a unique file name to include project name and data date using (yyyy-mm-dd) convention. Each native NAS must have a unique Project ID and Project Name.

- d. Log Report: P6 Scheduling/Leveling Report.

- e. Narrative Report: Identify and justify:

Each entry in the Narrative Report must cite the respective Activity ID and Activity Name, the date and reason for any changes, and description of the change.

- (1) Provide Project Summary Data in format below:

- (a) Data Date \_\_\_\_\_
  - (b) Award Date: \_\_\_\_\_
  - (c) Original Project Duration: \_\_\_\_\_ days post Award Date
  - (d) Current Project Duration: \_\_\_\_\_ days post Award Date
  - (e) Time percent elapsed: \_\_\_\_\_ percent at data date
  - (f) Original CCD: \_\_\_\_\_
  - (g) Current CCD: \_\_\_\_\_ (thru MOD \_\_\_\_\_)
  - (h) Anticipated CCD: \_\_\_\_\_ (\_\_\_\_ calendar days early/late)
  - (i) Original Contract Value: \$\_\_\_\_\_
  - (j) Current Contract Value: \$\_\_\_\_\_
  - (k) Invoiced Amount: \$\_\_\_\_\_ (\_\_\_\_ percent)
  - (l) Cost Growth: \_\_\_\_\_ percent
  - (m) NAS Growth: \_\_\_\_\_ percent
  - (n) There are a total of \_\_\_\_\_ activities, \_\_\_\_\_ activities complete (\_\_\_\_ percent), \_\_\_\_\_ activities in progress (\_\_\_\_ percent), \_\_\_\_\_ activities not started (\_\_\_\_ percent). Of the in progress and not started activities; \_\_\_\_\_ (\_\_\_\_ percent) are on the longest path. The longest path has duration of \_\_\_\_\_ calendar days from data-date to anticipated project completion.
- (2) Progress made in each area of the project;
  - (3) Longest Path;
  - (4) Date/time constraint(s), other than those required by the contract
  - (5) Listing of all changes made between the previous monthly NAS update and current monthly NAS update include as a minimum: added or deleted activities, original and remaining durations for activities that have not started, logic (sequence constraint lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading. Identify any other changes that relate to calculations within Oracle Primavera P6;
  - (6) Any decrease in previously reported activity Earned Amount;
  - (7) Pending items and status thereof, including permits, changes orders, and time extensions;
  - (8) Status of Contract Completion Date and interim milestones;
  - (9) Status of Projected Completion Milestone and account of difference in calendar days between previous update Projected

Completion Milestone

- (10) Current and anticipated delays listing Activity Names and IDs for impacted activities(describe cause of delay and corrective actions(s) and mitigation measures to minimize);
  - (11) Description of current and potential future NAS problem areas.
  - (12) Identification of any weather and restricted lost time as compared to anticipated weather for the month and anticipated restricted days for which the update is submitted. Impacts resulting from adverse weather must be documented in tabular form showing the calendar month (or billing period) with the days on which construction activity incurred Lost Work Days due to adverse weather. In narrative form, describe the adverse weather cause such as precipitation measurement, temperature, wind or other influencing factors, and why work was impacted. Identify the construction activity(s) by ID and Name that was (were) scheduled, impacted.
- f. Earned Value Report: Derive from and correspond to P6 cost loaded Monthly NAS Update. List all activities having a budget amount cost loaded. Compile total earnings on the project from notice to proceed to current progress payment request. Show on the report for each activity:
- (1) Current budget;
  - (2) Previous Physical Percent Complete;
  - (3) To-Date Physical Percent Complete;
  - (4) Previous Earned Value;
  - (5) To-Date Earned Value;
  - (6) Cost This Period;
  - (7) Cost to Complete.
- g. Schedule Variance Control (SVC) Diagram: With each NAS submission, provide a SVC diagram representing only the current file showing 1) A Cash Flow Curve indicating planned project cost based on each of projected early and projected late activity finish dates and 2) one curve for Earned Value to-date. Revise Cash Flow Curves when the contract is modified, or as directed by the Contracting Officer Include a legend on report clearly indication 3 curves: early finish, late finish, and earned-value to date.

Use the following settings in Activity Usage Profile Options:

- (1) In the Data section, under Display, the radio box for Cost must be selected.
- (2) In the Data section, under Filter for Bars/Graphs, the checkbox for Total must be checked.
- (3) In the Show Bars/Curves section:

- (a) Under the By Date column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for Budgeted and Remaining Early must be unchecked.
  - (b) Under the Cumulative column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for Budgeted and Remaining Early must be unchecked.
  - (c) Set the color for Baseline to green.
  - (d) Set the color for Actual to blue.
  - (e) Set the color for Remaining Late to red.
- (4) In the Show Earned Value Curves section, the checkboxes for Planned Value Cost, Earned Value Cost and Estimate at Completion must be unchecked.
- h. Logic Diagram showing timescale from data date to 60 days after data date with filter for longest path. Leave Group By selection blank and sort by finish date in ascending order.
- i. Logic Diagram showing timescale from data date to contract completion date with filter for longest path. Leave Group By selection blank and sort by finish date in ascending order.
- j. Baseline or Monthly Update Checklist as applicable completed and certified by Qualified Scheduler. Contractor Baseline Project Schedule Review, Contractor Monthly Update Schedule Review, and Contractor Baseline Project Schedule, and General Commissioning/HVAC Review Checklist. Checklists can be found on the Whole Building Design Guide website as [Related Materials](#) to UFGS 01 32 17.00 20.
- k. Screen shot PDF of Primavera P6 Time Periods Settings referenced in paragraph NAS SOFTWARE SETTINGS AND RESTRICTIONS, list item c.(2): ADMIN DROP-DOWN MENU, ADMIN PREFERENCES, TIME PERIODS TAB
- l. Screen shot PDF of Primavera P6 Earned Value Settings referenced in paragraph NAS SOFTWARE SETTINGS AND RESTRICTIONS, list item c.(3): ADMIN DROP-DOWN MENU, ADMIN PREFERENCES, EARNED VALUE TAB
- m. Daily Reported Production Activity: Submit on a monthly basis, in electronic spreadsheet, summary of daily reported production activity for the reporting month in the update NAS. Use the following columns for reporting:
  - (1) Date
  - (2) Activity ID
  - (3) Work Description
  - (4) Contractor
  - (5) Billable Hours

## 1.11 CONTRACT MODIFICATION

### 1.11.1 Time Impact Analysis (TIA)

Submit a Time Impact Analysis with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, impacts the longest path, and extends the Projected Completion beyond the Contract Completion Date.

a. Each TIA must be in both narrative and NAS form. The narrative must:

- (1) Define the scope and conditions of the change;
- (2) Provide start and finish dates of impact to longest path;
- (3) Identify and include any accepted NAS referenced in TIA and any changes made to accepted NAS to prepare additional NAS included in the TIA;
- (4) Successor(s) and predecessor(s) activities to impact period;
- (5) Responsible party;
- (6) Describe how the impact originated;

b. The TIA submission must consist of at least three native XER files:

- (1) Fragnet used to define the scope of the changed condition
- (2) Most recent accepted baseline NAS or monthly NAS update as of the time of the impact start date. If the data date of this NAS is different than the start date of impact, update this NAS to show all activity progress as of the time of the impact start. The impact start date is identified as the time when an existing activity is impeded for either starting or finishing. This NAS must have a data date that is the same as the identified start date of impact in "a.(2)". Identify any changes made to most recent accepted NAS.
- (3) The impacted NAS that has the Fragnet inserted in the updated NAS from "b.(2)" and the NAS scheduled using the Schedule command in Oracle Primavera P6 so that the new completion date is determined.

c. For claimed as-built project delay, the inserted Fragnet TIA method must be modified to account for as-built events known to occur after the data date of NAS update used. Updated NAS for periods following the impact start date will be used and included in submittal to evaluate how the project progressed (as-built) through the finish of impact. Impact to longest path must be determined for each following update period.

d. All TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events. The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the

earliest delay event.

(1) Identify and classify types of delay defined as follows:

- (a) Force majeure delay (e.g., weather delay): Any delay event caused by something or someone other than the Government or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, classified as a non-compensable event.
- (b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the contractor-delay is on the longest path, in absence of other types of concurrent delays, Contractor is not granted extension of contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, Contractor must provide to Contracting Officer a Corrective Action Plan identifying plan to mitigate delay.
- (c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.

(2) Functional concurrency must be used to analyze concurrent delays, where: separate delay issues delay project completion, do not necessarily occur at same time, rather occur within same monthly NAS update period at minimum, or within same as-built period under review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined in the following combinations:

- (a) Government-delay concurrent with contractor-delay: excusable time extension, classified non-compensable event.
- (b) Government-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.
- (c) Contractor-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(3) Pacing delay reacting to another delay (parent delay) equally or more critical than paced activity must be identified prior to pacing. Contracting Officer will notify Contractor prior to pacing. Contractor must notify Contracting Officer prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:

- (a) Government-delay concurrent with contractor-pacing: excusable time extension, classified compensable event.

(b) Contractor-delay concurrent with Government-pacing:  
inexcusable time extension, classified non-compensable event

- e. Submit electronic files containing the narrative and the Native XER  
NAS files used and or referenced in the time impact analysis.

#### 1.12 PROJECT FLOAT

Project Float is the length of time between the Project Completion  
Milestone and the Contract Completion Date. Project Float available in  
the NAS will not be for the exclusive use of either the Government or the  
Contractor.

The use of Resource Leveling is prohibited. Techniques used for the  
purpose of artificially adjusting activity durations to consume float and  
influence longest path are prohibited.

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

Not used.

-- End of Section --

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**08/22**

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DEMOLITION  
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO M 145 (1991; R 2012) Standard Specification for Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes

AASHTO T 180 (2017) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-460-01 (2019; with Change 3, 2023) Design: Petroleum Fuel Facilities

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61 National Emission Standards for Hazardous Air Pollutants

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

#### 1.2.1.2 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

#### 1.2.2 Demolition Plan

Prepare a [Demolition Plan](#) and submit proposed salvage, demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services, a detailed description of methods and equipment to be used for each operation and of the sequence of operations. [Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed.](#) [Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use.](#) Coordinate with Waste Management Plan in accordance with Section [01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL](#). Provide procedures for safe conduct of the work in accordance with [EM 385-1-1](#). Plan must be approved by Contracting Officer prior to work beginning.

#### 1.2.3 General Requirements

Do not begin demolition until authorization is received from the Contracting Officer. [The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials.](#) The work includes demolition, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with [EM 385-1-1](#), Section 23, Demolition, and other applicable Sections.

#### 1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

##### 1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

### 1.3.2 Trees

Protect trees within the project site which might be damaged during demolition, and which are indicated to be left in place, by a 1.8 m high fence. Erect and secure fence a minimum of 1.5 m from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

### 1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor.

### 1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

## 1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

## 1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

## 1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-01 Preconstruction Submittals

Demolition Plan; G

Existing Conditions

## 1.7 QUALITY ASSURANCE

Notify the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with

local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in **ASSP A10.6**. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

#### 1.7.1 Dust and Debris Control

Prevent the spread of dust and debris and avoid the creation of a nuisance in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, flooding, or pollution.

### 1.8 PROTECTION

#### 1.8.1 Traffic Control Signs

Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Anchor barricades in a manner to prevent displacement by wind. Notify the Contracting Officer prior to beginning such work.

#### 1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the site specific features or structure being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

### 1.9 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

#### 1.10 EXISTING CONDITIONS

Before beginning any demolition work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

## PART 2 PRODUCTS

### 2.1 FILL MATERIAL

- a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill voids, depressions or excavations resulting from demolition of structures. Provide fill material consisting of waste products from demolition until all waste appropriate for this purpose is consumed.
- b. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as follows:

|                            |                             |
|----------------------------|-----------------------------|
| Soil classification        | AASHTO M 145                |
| Moisture-density relations | AASHTO T 180, Method B or D |

## PART 3 EXECUTION

### 3.1 EXISTING FACILITIES TO BE REMOVED

Disassemble existing construction scheduled to be removed for reuse. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

#### 3.1.1 Structures

- a. Locate demolition equipment throughout the structure and remove materials so as to not impose excessive loads to supporting walls, floors, or framing.

#### 3.1.2 Utilities and Related Equipment

##### 3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

##### 3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

#### 3.1.3 Chain Link Fencing

Remove chain link fencing, gates and other related salvaged items

scheduled for removal and transport to designated areas. Remove gates as whole units. Cut chain link fabric to 7 m lengths and store in rolls off the ground.

#### 3.1.4 Paving and Slabs

Sawcut and remove concrete and asphaltic concrete paving and slabs including aggregate base as indicated. Move, grind and store pavement and slabs designated to be recycled and utilized in this project as directed by the Contracting Officer. Remove pavement and slabs not to be used in this project from the project site at Contractor's expense. If pavement remnant is 1 m or less, then remove and restore pavement.

#### 3.1.5 Concrete

Saw concrete along straight lines to a depth of a minimum 50 mm. Make each cut in walls perpendicular to the face and in alignment with the cut in the opposite face. Break out the remainder of the concrete provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or saw cut entirely through the concrete. Sawcuts in existing concrete sidewalk, roads, concrete pads, and parking areas must be at the nearest existing expansion joint or weakened plane joint and at full depth.

#### 3.1.6 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, steel trusses, metal gutters, roofing and siding, metal toilet partitions, toilet accessories and similar items. Recycle scrap metal as part of demolition operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

#### 3.1.7 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a manner to produce surfaces suitable for receiving new work. Make finished surfaces of patched area flush with the adjacent existing surface and match the existing adjacent surface as closely as possible to texture and finish. Provide patching as specified and indicated, and include the following:

- a. Concrete: Completely fill holes and depressions, caused by previous physical damage or left as a result of removals in existing concrete to remain, with an approved concrete material, applied in accordance with specifications.

#### 3.1.8 Mechanical Equipment and Fixtures

##### 3.1.8.1 Piping

Disconnect piping at unions, flanges and valves, and fittings as required

to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing. Classify piping not designated for salvage, or not reusable, as scrap metal.

#### 3.1.8.2 Fuel Piping

Drain, vacuum extract, pig (where possible), and purge fuel piping to be removed or abandoned in place in accordance with Chapter 14 of UFC 3-460-01.

#### 3.1.9 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

##### 3.1.9.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

##### 3.1.9.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

##### 3.1.9.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

##### 3.1.9.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

#### 3.1.10 Items With Unique/Regulated Disposal Requirements

Remove and dispose of items with unique or regulated disposal requirements

in the manner dictated by law or in the most environmentally responsible manner.

### 3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition work in areas occupied by structures to be demolished until all demolition in the area has been completed and debris removed. Fill holes and other hazardous openings.

### 3.3 DISPOSITION OF MATERIAL

#### 3.3.1 Title to Materials

Except for salvaged items, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

#### 3.3.2 Reuse of Materials and Equipment

Remove and store materials and equipment to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

#### 3.3.3 Salvaged Materials and Equipment

Remove materials and equipment that are indicated to be salvaged or are to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site.

- a. Salvage items and material to the maximum extent possible.
- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the diversion calculations for the project.
- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers.
- d. Remove the following items reserved as property of the using service prior to commencement of work under this contract: signs, concrete, barriers, vinyl fencing, bus stop shelter, and guardrail.

### 3.3.4 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible and noncombustible material per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

### 3.4 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

### 3.5 DISPOSAL OF REMOVED MATERIALS

#### 3.5.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable local regulations as contractually specified.

#### 3.5.2 Burning on Government Property

Burning of materials removed from demolished structures will not be permitted on Government property.

#### 3.5.3 Removal to Spoil Areas on Government Property

Transport noncombustible materials removed from demolished structures to designated spoil areas on Government property.

#### 3.5.4 Removal from Government Property

Transport waste materials removed from demolished structures, except waste soil, from Government property for legal disposal. Dispose of waste soil per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

### 3.6 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

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CEMENT STUCCO  
08/17, CHG 2: 11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A1064/A1064M | (2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete                                                    |
| ASTM C150/C150M   | (2024) Standard Specification for Portland Cement                                                                                                                      |
| ASTM C206         | (2014; R 2022) Standard Specification for Finishing Hydrated Lime                                                                                                      |
| ASTM C897         | (2015; R 2020) Aggregate for Job-Mixed Portland Cement-Based Plasters                                                                                                  |
| ASTM C926         | (2024) Standard Specification for Application of Portland Cement-Based Plaster                                                                                         |
| ASTM D1784        | (2025) Standard Classification System and Basis for Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds |

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Proportions and Mixing

SD-04 Samples

Sample Panel; G

1.3 QUALITY ASSURANCE

Submit a SAMPLE PANEL as follows: A sample panel of stucco, constructed

at the jobsite, and located as directed, to demonstrate installation procedures, texture and color, prior to proceeding with any stucco work; panel size must be a minimum of 1200 mm wide x 2400 mm high; containing each type accessory proposed for use and constructed in the vertical position. Sample panel must have exposed reinforcement at the edges. Each phase of installation such as framing, scratch coat, brown coat, finish coat and curing procedures must be demonstrated in the construction of the panel. Submit one 300 mm square of reinforcement and one 300 mm length of each accessory proposed for use, prior to constructing the sample panel.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver packaged materials to the site in the original packages and containers with labels intact and seals unbroken. Keep cementitious materials dry and stored off the ground, under cover and away from damp surfaces until ready to be used. Aggregate must be covered to prevent the absorption or loss of moisture.

#### 1.5 ENVIRONMENTAL REQUIREMENTS

Do not apply stucco when the ambient temperature is 4 degrees C or lower, or when a drop in temperature below 4 degrees C is expected within 48 hours after application.

### PART 2 PRODUCTS

#### 2.1 PORTLAND CEMENT

Portland cement must conform to ASTM C150/C150M, gray Portland cement Type I.

#### 2.2 LIME

Lime must conform to ASTM C206, Type S.

#### 2.3 SAND

Sand aggregate for job-mixed base coat and job-mixed finish coat stucco must conform to ASTM C897.

#### 2.4 ACCESSORIES

Accessories must be roll formed galvanized steel, except that cornerite and striplath must be formed from steel sheets with manufacturer's standard galvanized coating. Vinyl members must be in accordance with ASTM D1784. Welded wire corner reinforcements must be zinc coated, galvanized 1.4 mm steel wire conforming to ASTM A1064/A1064M. Furring must include hangers, bolts, inserts, clips, fastenings, and attachments of number, size, and design to develop the full strength of the members.

#### 2.5 WATER

Provide clean, fresh, potable water, free from amounts of oils, acids, alkalis and organic matter that would be injurious to the stucco.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

### 3.2 PREPARATION OF SURFACES

Preparation of surfaces for application of stucco to solid bases such as stone, masonry or concrete must conform to the applicable requirements of [ASTM C926](#).

### 3.3 PROPORTIONS AND MIXING

Proportions and mixing for job-mixed base coat and finish coat must conform to the applicable requirements of [ASTM C926](#). Mixing of mill-mixed finish coat must be in accordance with the manufacturer's directions. Submit detailed description of the proposed job-mix proportions for base and finish coats; including identification of thickness of coats.

### 3.4 STUCCO APPLICATION

Stucco must be applied in three coats to a thickness of not less than [25 mm](#) as measured from the back plane of metal reinforcement, exclusive of ribs or dimples or from the face of solid backing or support, with or without metal reinforcement, to the finished stucco surface, including moderate texture variations. Stucco application must conform to the applicable requirements of [ASTM C926](#) and the following:

#### 3.4.1 Workmanship

Items or features of the work in connection with or adjoining the stucco must be in place, plumb, straight, and true prior to beginning the stucco work. Metal and wire lath, where required, must be in place and positioned to provide a good key at back of lath. Where lath is applied over copper, the copper must be given a heavy coat of bituminous paint. Masonry surfaces to receive stucco must be evenly dampened immediately prior to application of stucco. Each stucco coat must be applied continuously in one general direction, without allowing mortar to dry at edges. Where it is impossible to work the full dimension of a wall surface in a continuous operation, jointing must be made at a break, opening, or other natural division of the surface. Edges to be joined must be dampened slightly to produce a smooth confluence. Exterior corners of stucco must be slightly rounded. Stucco on soffit surfaces must be pitched forward to form a drip.

#### 3.4.2 Scratch Coat

Apply scratch coat not less than [10 mm](#) thick under sufficient pressure to form good keys and to completely embed the reinforcement. Before the scratch coat has set, it must be lightly scratched in one direction and vertical surfaces must be scratched in the horizontal direction only. The scratch coat must be fog cured for a minimum of 72 hours.

#### 3.4.3 Brown Coat

Evenly dampen the scratch coat to obtain uniform suction before the brown coat is applied. There must be no visible water on the surface when the

brown coat is applied. The brown coat must be applied to the scratch coat with sufficient pressure to force the stucco into the scratches and must be brought to a plumb, true, even plane with rod or straightedge. When set sufficiently, the brown coat must be uniformly floated with a dry float to promote densification of the coat and to provide a surface receptive to bonding of the finish coat. Brown coat must be fog cured for a minimum of 72 hours.

#### 3.4.4 Finish Coat

Dampen surfaces of the brown coat not more than 1 hour before the finish coat is to be applied to a uniform wetness with no free-standing water on the surface. The finish coat must have a smooth trowel finish and must conform to the approved sample. Fog cure the finish coat for a minimum of 48 hours. Take care to prevent staining.

#### 3.4.5 Surface Tolerance

When a 3 m straightedge is placed at any location on the finished surface of the stucco, excluding rough-textured finish, the surface must not vary more than 3 mm from the straightedge.

### 3.5 CURING AND PROTECTION

Perform fog curing by applying a fine mist of water to the stucco. Exercise care during fog curing to avoid erosion damage of the stucco surfaces. Do not use a solid stream of water. Fog not less than three times daily. Protect the stucco from the direct rays of the sun during severe drying conditions using canvas, cloth or other approved sheet material.

### 3.6 PATCHING AND POINTING

Replace or patch loose, cracked, damaged or defective work as directed. Patching must match existing work in texture and color and must be finished flush.

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EXTERIOR SIGNAGE  
08/17, CHG 1: 11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS C1.1M/C1.1 (2019) Recommended Practices for Resistance Welding

AWS D1.1/D1.1M (2025) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A653/A653M (2025) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A924/A924M (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process

ASTM B26/B26M (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings

ASTM B108/B108M (2019) Standard Specification for Aluminum-Alloy Permanent Mold Castings

ASTM B209M (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)

ASTM B221M (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500 (2006) Metal Finishes Manual

1.2 GENERAL REQUIREMENTS

All exterior signage must be provided by a single manufacturer. Exterior signage must be of the design, detail, sizes, types, and message content shown on the drawings, must conform to the requirements specified, and

must be provided at the locations indicated. Submit exterior signage schedule in electronic media with spread sheet format. Spread sheet must include sign location, sign type, and message. Signs must be complete with lettering, framing as detailed, and related components for a complete installation. Each sample must consist of a complete sign panel with letters and symbols. Samples may be installed in the work, provided each sample is identified and location recorded. Submit three color samples for each material requiring color and 305 mm square sample of sign face color sample.

#### 1.2.1 Wind Load Requirements

Exterior signage must be designed to withstand 177 km/h windload. Submit design analysis and supporting calculations performed in support of specified signage.

#### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Approved Detail Drawings; G

##### SD-03 Product Data

Exterior Signage System

Installation

Exterior Signage; G

Wind Load Requirements

##### SD-04 Samples

Exterior Signage; G

##### SD-10 Operation and Maintenance Data

Protection and Cleaning; G

#### 1.4 QUALIFICATIONS

Signs, plaques, and dimensional letters must be the standard product of a manufacturer regularly engaged in the manufacture of the products. Items of equipment must essentially duplicate equipment that has been in satisfactory use at least 2 years prior to bid opening.

#### 1.5 DELIVERY AND STORAGE

Materials must be wrapped for shipment and storage, delivered to the jobsite in manufacturer's original packaging, and stored in a clean, dry area in accordance with manufacturer's instructions.

## 1.6 WARRANTY

Manufacturer's standard performance guarantees or warranties that extend beyond a one year period must be provided.

## PART 2 PRODUCTS

### 2.1 EXTERIOR SIGNAGE SYSTEM

Exterior signage must consist of identification and regulatory type signs located where shown. Dimensions, details, materials, message content, and design of signage must be as shown. Submit manufacturer's descriptive data and catalog cuts.

#### 2.1.1 Panel And Post/Panel Type Signs

##### 2.1.1.1 Panels

Message panels must be provided in sizes shown on drawings. Panels must be fabricated a minimum of 3.2 mm aluminum.

##### 2.1.1.2 Finishes

Metal panel system finish must be baked enamel or two-component acrylic polyurethane.

### 2.2 GRAPHICS FOR EXTERIOR SIGNAGE SYSTEMS

#### 2.2.1 Graphics

Signage graphics must conform to the following:

- a. Message must be applied using the frisket method. Photomechanically reproduced graphic masks must be applied to the sign face which has been coated with the graphics color. A background must then be applied to the exposed surfaces. Handcut masks will not be accepted. Edges that are nicked, cut, or ragged will not be acceptable. A protective overcoat containing UV-resistant additives must be applied.

#### 2.2.2 Messages

See drawings for message content. Typeface: base standard. Type size as indicated.

### 2.3 ALUMINUM ALLOY PRODUCTS

Aluminum alloy products must conform to ASTM B209M for sheet or plate, ASTM B221M for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings. Aluminum extrusions must be provided at least 3 mm thick and aluminum plate or sheet at least 16 gauge thick. Welding for aluminum products must conform to AWS C1.1M/C1.1.

### 2.4 ORGANIC COATING

Clean, prime and give surfaces a semi-gloss baked enamel or two-component acrylic polyurethane finish in accordance with NAAMM AMP 500, AMP 505, with total dry film thickness not less than 0.030 mm.

## 2.5 ANCHORS AND FASTENERS

Exposed anchor and fastener materials must be compatible with metal to which applied and must match in color and finish and must be non-rusting, non-corroding, and non-staining. Exposed fasteners must be tamper-proof.

## 2.6 SHOP FABRICATION AND MANUFACTURE

### 2.6.1 Factory Workmanship

Work must be assembled in the shop, as far as practical, ready for installation at the site. Work that cannot be shop assembled must be given a trial fit in the shop to ensure proper field assembly. Holes for bolts and screws must be drilled or punched. Drilling and punching must produce clean, true lines and surfaces. Welding to or on structural steel must be in accordance with [AWS D1.1/D1.1M](#). Welding must be continuous along the entire area of contact. Exposed welds must be ground smooth. Exposed surfaces of work must have a smooth finish and exposed riveting must be flush. Fastenings must be concealed where practical. Items specified to be galvanized must be by hot-dip process after fabrication if practical. Galvanization must be in accordance with [ASTM A123/A123M](#) and [ASTM A653/A653M](#), as applicable. Other metallic coatings of steel sheet must be in accordance with [ASTM A924/A924M](#). Joints exposed to the weather must be formed to exclude water. Drainage and weep holes must be included as required to prevent condensation buildup.

### 2.6.2 Dissimilar Materials

Where dissimilar metals are in contact, or where aluminum is in contact with concrete, mortar, masonry, wet or pressure-treated wood, or absorptive materials subject to wetting, the surfaces must be protected with a coat of asphalt varnish or a coat of zinc-molybdate primer to prevent galvanic or corrosive action.

### 2.6.3 Shop Painting

Surfaces of miscellaneous metal work, except nonferrous metal, corrosion resisting steel, and zinc-coated work, must be given one coat of zinc-molybdate primer or an approved rust-resisting treatment and metallic primer in accordance with manufacturer's standard practice. Surfaces of items to be embedded in concrete must not be painted. Upon completion of work, damaged surfaces must be recoated.

## 2.7 COLOR, FINISH, AND CONTRAST

Color must be as indicated on the drawings. For buildings required to be handicapped-accessible, the characters and background of signs must be eggshell, matte, or other non-glare finish. Characters and symbols must contrast with their background - either light characters on a dark background or dark characters on a light background.

## PART 3 EXECUTION

### 3.1 [INSTALLATION](#)

Signs, plaques, or dimensional letters must be installed in accordance with approved manufacturer's instructions at locations shown on the [approved detail drawings](#); submit drawings showing elevations of each type of sign; dimensions, details, and methods of mounting or anchoring; shape

and thickness of materials; and details of construction. A schedule showing the location, each sign type, and message must be included. Circuits installed underground must conform to the requirements of Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Steel conduits installed underground and illuminated signage mounted directly on buildings must be in conformance with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Signs must be installed plumb and true at mounting heights indicated, and by method shown or specified. Signs mounted on other surfaces must not be installed until finishes on such surfaces have been completed. Submit manufacturer's installation instructions and cleaning instructions.

#### 3.1.1 Anchorage

Anchorage and fastener materials must be in accordance with approved manufacturer's instructions for the indicated substrate. Anchorage not otherwise specified or indicated must include slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; lag bolts and screws for wood.

#### 3.1.2 Protection and Cleaning

The work must be protected against damage during construction. Hardware and electrical equipment must be adjusted for proper operation. Glass, frames, and other sign surfaces must be cleaned in accordance with manufacturer's instructions. After signs are completed and inspected, cover all project identification, directional, and other signs which may mislead the public. Covering must be maintained until instructed to be removed by the Contracting Officer or until the facility is to be opened for business. Submit six copies of maintenance instructions listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guides. The instructions must include simplified diagrams for the equipment as installed. Signs must be cleaned, as required, at time of cover removal.

#### 3.2 FIELD PAINTED FINISH

Miscellaneous metals and frames must be field painted in accordance with Section 09 90 00 PAINTS AND COATINGS. Anodized metals, masonry, and glass must be protected from paint. Finish must be free of scratches or other blemishes.

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GEONET  
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|            |                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------|
| ASTM D1505 | (2018) Standard Test Method for Density of Plastics by the Density-Gradient Technique                  |
| ASTM D1603 | (2020) Carbon Black Content in Olefin Plastics                                                         |
| ASTM D4218 | (2020) Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique |
| ASTM D5035 | (2011) Breaking Force and Elongation of Textile Fabrics (Strip Method)                                 |
| ASTM D5199 | (2012) Measuring Nominal Thickness of Geosynthetics                                                    |

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Sampling and Testing

Penetrations

Construction Quality Control (QC) Laboratory

SD-04 Samples

Geonet

Seams and Overlaps

SD-06 Test Reports

Sampling and Testing

Geonet

### 1.3 QUALITY ASSURANCE

Provide a **construction quality control (QC) laboratory** that has also performed quality assurance (QA) testing, if required, of geosynthetic drainage layers for at least five completed projects, having a total minimum area of **186,000 square meters**. Submit qualifications of laboratory carrying current accreditation via the Geosynthetic Accreditation Institute's Laboratory Accreditation Program (GAI-LAP) for the tests it will be required to perform.

### 1.4 DELIVERY, STORAGE, AND HANDLING

The QC inspector must be present during delivery and unloading of the **geonet**. Ensure the **geonet** material has not been damaged during shipping, storage, or handling. Repair or replace any **geonet** material found to be damaged. Accept delivery of material only after the required submittals have been approved. Label each roll with the manufacturer's name, product identification, lot number, roll number, and roll dimensions. Store the rolls in a level and dry area.

## PART 2 PRODUCTS

### 2.1 GEONET

Use polyethylene polymer, which is clean and free of any foreign contaminants, to manufacture the **geonet**. Submit one properly identified **610 by 610 mm** minimum size **geonet** sample; fasteners proposed for use; and the method of seaming and overlapping. Submit manufacturer's quality control test results. Use regrind material which consists of edge trimmings and other scraps to manufacture the **geonet**; however, do not use post-consumer recycled materials. Conform the **geonet** to the property requirements listed in Table 1. Component criteria for the **geonet** alone are also listed in Table 1. Where applicable, Table 1 property values represent minimum average roll values (MARV). The value for AOS represents the maximum average roll value (MaxARV).

| TABLE 1 - <b>GEONET</b>                                                                                                                                            |                                  |                 |                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------|-------------------------------|
| PROPERTY                                                                                                                                                           | TEST METHOD                      | TEST VALUE      | MINIMUM MQC TESTING FREQUENCY |
| <b>GEONET</b>                                                                                                                                                      |                                  |                 |                               |
| Thickness, minimum avg, Note 1                                                                                                                                     | <b>ASTM D5199</b>                | <b>7 mm</b>     | <b>9300 sq m</b>              |
| Polymer Density, minimum avg                                                                                                                                       | <b>ASTM D1505</b>                | 0.940 g/cc      | <b>9300 sq m</b>              |
| Carbon Black Content                                                                                                                                               | <b>ASTM D1603<br/>ASTM D4218</b> | 1-3 percent     | <b>9300 sq m</b>              |
| Tensile Strength, minimum avg, Note 2                                                                                                                              | <b>ASTM D5035</b>                | <b>7884 N/m</b> | <b>9300 sq m</b>              |
| Note 1: The diameter of the presser foot must be <b>56 mm</b> and the pressure must be <b>20 kPa</b> . For other thickness options, see manufacturer's literature. |                                  |                 |                               |
| Note 2: This is the average peak value for five equally spaced machine direction tests across the roll width.                                                      |                                  |                 |                               |

## 2.2 SAMPLING AND TESTING

### 2.2.1 Manufacturing Quality Control Testing

Provide manufacturing quality control test methods and frequencies in accordance with Table 1 unless otherwise approved. Submit manufacturer's quality control manual and construction quality control test results.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Surface Preparation

Prior to placement of the geonet, provide a subgrade that is smooth and free of all materials which could damage the geonet.

#### 3.1.2 Placement

Do not damage the geonet during placement. Unroll the geonet, keeping the net flat against the subgrade to minimize wrinkles and folds. Do not drag the drainage layer across textured surfaces.

#### 3.1.3 Seams and Overlaps

##### 3.1.3.1 Geonet Side Seams

Overlap geonet side seams a minimum of 100 mm. Space side seam fasteners a maximum of 1.5 m. In anchor trenches, space fasteners a maximum of 305 mm.

##### 3.1.3.2 Geonet End Seams

Overlap geonet end seams a minimum of 305 mm. Space end seam fasteners spacing a maximum of 305 mm. Make overlaps in the direction of flow.

##### 3.1.3.3 Geonet Fasteners

Tie geonet rolls together with plastic fasteners. Use fasteners that are a contrasting color from the geonet and attached geotextiles. Metallic fasteners will not be allowed.

#### 3.1.4 Penetrations

Submit penetration details.

### 3.2 REPAIRS

#### 3.2.1 Geonet Damage

Make repairs by placing a patch of the geonet over the damaged area. Extend the patch a minimum of 610 mm beyond the edge of the damage. Use approved fasteners, spaced every 150 mm around the patch, to hold the patch in place. If more than 25 percent of the roll width is damaged, approval must be obtained to repair or replace the damaged roll.

### 3.3 PROTECTION AND BACKFILLING

Cover the geonet with the specified materials within 14 days of

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NAVSTA ROTA, SPAIN

acceptance. Do not operate equipment on the top surface of the geonet without permission from the Contracting Officer.

-- End of Section --

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SECTION 33 52 40

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FUEL SYSTEMS PIPING (NON-HYDRANT)  
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PART 1 GENERAL

1.1 SUMMARY

This section defines the requirements for pipe, piping components, and valves as related to an non-hydrant, non-service station, fuel distribution system. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, and other miscellaneous piping systems. Provide the entire fuel distribution system as a complete and fully operational system. Size, select, construct, and install equipment and system components to operate together as a complete system. Substitutions of functions specified herein will not be acceptable. Coordinate the work of the system manufacturer's service personnel during construction, testing, calibration, and acceptance of the system. System components and piping associated with the F-24 storage tank system must be designed to handle a working pressure of 1965 kPa (285 psi) at 38 deg C with a corresponding Class 150 Flange Rating. The JP-5 Pipeline from Pier '3' to Pit 'ML-4' must be designed for a working pressure of 2520 kPa (365 psi) at 38 deg C with a corresponding Class 300 Flange Rating. Components specified herein must be compatible with the fuel to be handled. Components to be suitable for outside, unsheltered location, and to function normally in ambient temperatures.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

|             |                                                                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API RP 540  | (1999; R 2004) Electrical Installations in Petroleum Processing Plants                                                                                                                      |
| API RP 1110 | (2013; R 2018) Recommended Practice for the Pressure Testing of Steel Pipelines for the Transportation of Gas, Petroleum Gas, Hazardous Liquids, Highly Volatile Liquids, or Carbon Dioxide |
| API RP 1595 | (2012; R 2019; 2nd Ed) Design, Construction, Operation, Maintenance, and Inspection of Aviation Pre-Airfield Storage Terminals                                                              |
| API RP 2003 | (2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents                                                                                            |
| API RP 2009 | (2002; R 2007; 7th Ed) Safe Welding, Cutting, and Hot Work Practices in                                                                                                                     |

Refineries, Gasoline Plants, and  
Petrochemical Plants

|              |                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------|
| API RP 2200  | (1999; R 2004) Electrical Installations in Petroleum Processing Plants                                     |
| API STD 600  | (2015) Steel Gate Valves-Flanged and Butt-welding Ends, Bolted Bonnets                                     |
| API STD 608  | (2012) Metal Ball Valves - Flanged, Threaded, And Welding End                                              |
| API Spec 5L  | (2018; 46th Ed; ERTA 2018) Line Pipe                                                                       |
| API Spec 6D  | (June 2018, 4th Ed; Errata 1 July 2018; Errata 2 August 2018) Specification for Pipeline and Piping Valves |
| API Spec 6FA | (1999; R 2006; Errata 2006; Errata 2008; R 2011) Specification for Fire Test for Valves                    |
| API Std 607  | (2016) Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats                       |

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                       |                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| ASME B1.1             | (2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)                                                     |
| ASME B16.5            | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard                                   |
| ASME B16.9            | (2018) Factory-Made Wrought Buttwelding Fittings                                                                       |
| ASME B16.11           | (2016) Forged Fittings, Socket-Welding and Threaded                                                                    |
| ASME B16.21           | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                                                                       |
| ASME B18.2.1          | (2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)                                                      |
| ASME B18.2.2          | (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series) |
| ASME B31.3            | (2020) Process Piping                                                                                                  |
| ASME BPVC SEC VIII D1 | (2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1                                         |

ASTM INTERNATIONAL (ASTM)

|               |                                                           |
|---------------|-----------------------------------------------------------|
| ASTM A36/A36M | (2019) Standard Specification for Carbon Structural Steel |
|---------------|-----------------------------------------------------------|

|                 |                                                                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A53/A53M   | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                                          |
| ASTM A105/A105M | (2021) Standard Specification for Carbon Steel Forgings for Piping Applications                                                                                |
| ASTM A182/A182M | (2021) Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service  |
| ASTM A193/A193M | (2020) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications        |
| ASTM A194/A194M | (2020a) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both        |
| ASTM A234/A234M | (2019) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service                            |
| ASTM A269/A269M | (2015; R 2019) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service                                            |
| ASTM A312/A312M | (2021) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes                                                   |
| ASTM A358/A358M | (2019) Standard Specification for Electric-Fusion-Welded Austenitic Chromium-Nickel Stainless Steel Pipe for High-Temperature Service and General Applications |
| ASTM A403/A403M | (2020) Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings                                                                           |
| ASTM A961/A961M | (2021) Standard Specification for Common Requirements for Steel Flanges, Forged Fittings, Valves, and Parts for Piping Applications                            |
| ASTM B117       | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                                                              |
| ASTM B696       | (2000; R 2015) Standard Specification for Coatings of Cadmium Mechanically Deposited                                                                           |
| ASTM B766       | (1986; R 2015) Standard Specification for Electrodeposited Coatings of Cadmium                                                                                 |
| ASTM D229       | (2019) Standard Test Methods for Rigid                                                                                                                         |

Sheet and Plate Materials Used for  
Electrical Insulation

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| ASTM D1418 | (2010; R 2016) Standard Practice for Rubber and Rubber Lattices - Nomenclature                                      |
| ASTM D1655 | (2018a) Standard Specification for Aviation Turbine Fuels                                                           |
| ASTM D3308 | (2012; R 2017) Standard Specification for PTFE Resin Skived Tape                                                    |
| ASTM F336  | (2002; R 2016) Standard Practice for Design and Construction of Nonmetallic Enveloped Gaskets for Corrosive Service |
| ASTM F436  | (2011) Hardened Steel Washers                                                                                       |

BRITISH STANDARDS INSTITUTE (BSI)

|                 |                                                         |
|-----------------|---------------------------------------------------------|
| BS EN ISO 10497 | (2010) Testing of Valves Fire Type-Testing Requirements |
|-----------------|---------------------------------------------------------|

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| IEEE 142      | (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book |
| IEEE 1100     | (2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment                       |
| IEEE C62.41.2 | (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits        |

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

|          |                                          |
|----------|------------------------------------------|
| ISO 1629 | (2013) Rubber and Latexes - Nomenclature |
|----------|------------------------------------------|

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| MSS SP-58 | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |
|-----------|----------------------------------------------------------------------------------------------------------------|

NACE INTERNATIONAL (NACE)

|             |                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------|
| NACE SP0188 | (1999; R 2006) Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates |
|-------------|----------------------------------------------------------------------------------------------------|

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|         |                                                                   |
|---------|-------------------------------------------------------------------|
| NFPA 30 | (2021; TIA 20-1; TIA 20-2) Flammable and Combustible Liquids Code |
|---------|-------------------------------------------------------------------|

|          |                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------|
| NFPA 70  | (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4)<br>National Electrical Code |
| NFPA 77  | (2014) Recommended Practice on Static Electricity                                                          |
| NFPA 407 | (2022) Standard for Aircraft Fuel Servicing                                                                |
| NFPA 780 | (2020) Standard for the Installation of Lightning Protection Systems                                       |

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

|                      |                                   |
|----------------------|-----------------------------------|
| SSPC SP 5/NACE No. 1 | (2007) White Metal Blast Cleaning |
|----------------------|-----------------------------------|

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

|             |                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------|
| SAE AMS3275 | (2009; Rev C) Sheet, Acrylonitrile Butadiene (NBR) Rubber and Non-Asbestos Fiber Fuel and Oil Resistant |
| SAE J514    | (2012) Hydraulic Tube Fittings                                                                          |

U.S. DEPARTMENT OF DEFENSE (DOD)

|                |                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| MIL-DTL-5624   | (2016; Rev W; Notice 1 2020) Turbine Fuel, Aviation, Grades JP-4 and JP-5                                                      |
| MIL-PRF-13789  | (1999; Rev E; Notice 1 2008; Notice 2 1016; Notice 3 2021) Strainers, Sediment: Pipeline, Basket Type                          |
| MIL-STD-161    | (2005; Rev G; Notice 1 2010)<br>Identification Methods for Bulk Petroleum Products Systems Including Hydrocarbon Missile Fuels |
| MIL-STD-3004-1 | (2018) Quality Assurance for Bulk Fuels, Lubricants and Related Products                                                       |

### 1.3 ADMINISTRATIVE REQUIREMENTS

Design conditions must be as specified in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT. Refer to Section 01 78 23 OPERATION MAINTENANCE DATA.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

Pigging Plan; G

Hydrotesting Plan; G

Quality Assurance Plan; G

SD-02 Shop Drawings

Grounding and Bonding

Pipe Supports; G

Pigging System Components; G

SD-03 Product Data

Carbon Steel Piping; G

Stainless Steel Piping; G

External Protective Coatings For Aboveground Piping; G

External Protective Coatings For Buried Steel Piping; G

Fittings; G

Carbon Steel Fittings; G

Insulating Flange Kits; G

Lightning Surge Arrester; G

Bolts, Nuts And Washers; G

Ball Valves; G

Full Port Ball (DBBV) Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator

Swing Check Valves

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Pipe Sleeves; G

Strainers; G

Sample Connections; G

Sight Flow Indicators; G

Buried Utility Tape; G

Pipeline Markers; G

Flange Gaskets, Non-Metallic, Non-Electrically Isolating; G

SD-06 Test Reports

Exterior Coating Holiday Test

Pneumatic Test

Hydrostatic Test

Performance Testing

Pipe Pigging Verification

SD-07 Certificates

Contractor Qualifications; G

System Supplier; G

Pipeline Inventory; G

Qualifications of Owner's Inspector; G

Survey Final Elevations; G

Demonstrations

SD-10 Operation and Maintenance Data

Insulating Flange Kits; G

Lightning Surge Arrester; G

Ball Valves; G

Full Port Ball (DBBV) Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator; G

Swing Check Valves; G

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Strainers; G

Sample Connections; G

Sight Flow Indicators; G

1.5 QUALITY ASSURANCE

1.5.1 Contractor Qualifications

Each installation Contractor must have successfully completed at least 3

projects of the similar scope and the same size or larger within the last 6 years. Each installation Contractor must demonstrate specific installation experience in regard to the specific system installation to be performed. Each installation Contractor must have taken, if applicable, manufacturer's training courses on the installation of piping and must meet the licensing requirements in the state. Experience must include the erection of piping systems in compliance with the requirements of ASME B31.3, and NFPA 30. Submit a letter listing prior projects, the date of construction, a point of contact for each prior project, the scope of work of each prior project, and a detailed list of work performed. Provide in the letter evidence of prior manufacturer's training and state licensing.

#### 1.5.2 System Supplier

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

#### 1.5.3 Work Plan

Submit a comprehensive work plan that provides sufficient detail to demonstrate a thorough understanding of the project. Document that all components to be provided will function together and produce the result expected by the Government. Include any proposed dates for piping system shutdowns as well as the Contractor's ability to complete the work within the allotted shutdown periods. Show proposed dates and nature of piping system operations required of the Government. Include a list of manpower, spare piping and system component that will be on hand for each phase of the work. Describe, in detail, the means of:

- a. Coordinating work with Government and third parties.
- b. Preparing for safe piping repair work.
- c. Pneumatic pressure testing new piping sections.
- d. Hydrostatic pressure testing new piping sections.
- e. Interrupting or isolating an existing fuel service or system.
- f. Purging piping.
- g. Vapor monitoring.
- h. Preparations for containing and disposing of residual fuel.
- i. Cutting, sealing, and welding into existing piping systems.
- j. Welding tie-ins in place.
- k. Examining repair section tie-in welds.
- l. Collecting, storing and disposing of waste fuel generated during work.

#### 1.5.4 Pigging Plan (JP-5 Pipeline)

The Contractor must submit a detailed written plan covering all aspects of the pipeline pigging operation, including anticipated pig runs, types of pigs, sequence of work, and retrieval/repair procedures. The Contractor

must identify for each pigging evolution, the characteristics of the pig (type/purpose) and method/medium for propulsion.

#### 1.5.5 Hydrotesting Plan

The Contractor must submit a detailed written plan covering all aspects of the pipeline hydrostatic testing operations, including procedures and sequencing of testing, segments of piping to be tested, fluid to be used during testing, equipment removal and isolation, hydrostatic testing pressures, safety protocols, etc.

#### 1.5.6 Design Data

##### 1.5.6.1 Pipeline Inventory

Fuel system volume must be calculated using as constructed pipe lengths, internal diameters, fittings, and components. Totals must be provided for all items containing fuel with the exception of tanks which is covered by other specifications. A certified pipeline inventory - a detailed list with sizes, lengths, quantity, and volumes must be provided for the systems in this project. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, and other miscellaneous piping systems.

#### 1.5.7 Material and System Components Qualifications

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

#### 1.5.8 Nameplates

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

#### 1.5.9 Delivery, Storage, and Handling

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT. If fuel is used for hydrotesting or cleaning pigging, comply with all the requirements in Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

### PART 2 PRODUCTS

#### 2.1 ELECTRICAL WORK

##### 2.1.1 General

Motors, manual or automatic motor control system components except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

##### 2.1.2 Grounding and Bonding

Ground and bond as indicated on the drawings and in accordance with NFPA 70,

NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

## 2.2 MATERIALS

Pipe and fittings in contact with fuel must be stainless steel, interior epoxy coated carbon steel, or interior uncoated carbon steel. Fuel piping must be interior uncoated carbon steel, unless noted otherwise. No zinc coated metals, brass, bronze or other copper bearing alloys must be used in contact with the fuel. All carbon steel and stainless steel underground piping must have an exterior protective coating and must be cathodically protected in accordance with Section 26 42 17 IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM. Cathodic protection for metal components that attach to a tank must be coordinated and compatible with the tank corrosion control system. Identification of piping must be in accordance with MIL-STD-161 unless specified otherwise. Material for manual valves must be as specified hereinafter. Do not use aluminum valves.

### 2.2.1 Types of Fuel

Components must be suitable for use with F-24 turbine fuel (Jet A with additives FSII, CI/LE, and SDA); specific gravity 0.81 at 16 degrees C; viscosity 1.62 CS at 16 degrees C; Reid vapor pressure less than 0.35 kPa, ASTM D1655 or JP-5 turbine fuel; specific gravity 0.82 at 16 degrees C; viscosity 1.62 CS at 16 degrees C; Reid Vapor pressure less than 0.35 kPa, MIL-DTL-5624.

### 2.2.2 Carbon Steel Piping

Subject each length of pipe to factory hydrostatic testing and ultrasonic testing in accordance with their respective pipe specification.

- a. Piping 300 mm and Larger: API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless; or ASTM A53/A53M Grade B, seamless; "Standard" schedule wall thickness (0.375 in.; 9.525 mm).
- b. Piping 65 through 250 mm: Schedule 40, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Schedule 40, Seamless, ASTM A53/A53M Grade B.
- c. Piping 50 mm and Smaller: Schedule 80, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Seamless, Schedule 80 ASTM A53/A53M Grade B.
- d. For JP-5 pipeline system, HDD piping segments - The lowest 100 meter of each HDD pipe run must be internally factory coated with an epoxy coating in accordance with MIL-PRF-4556. The ends of the pipe must be masked or wiped back a minimum of 25 mm but not more than 40 mm.

### 2.2.3 Stainless Steel Piping

- a. Piping:
  - (1) ASTM A358/A358M, Grade 304L, Class 1 or Class 3 with supplementary requirements of S1, S2 and S3, or ASTM A312/A312M Type 304L, seamless (only). Any agreements between the purchaser and the manufacturer or supplier as referenced in the applicable ASTM must

include the Contracting Officer as a party to the agreement. All longitudinal piping welds will receive 100 percent radiographic inspection, 100 percent liquid penetrant inspection, 100 percent visual inspection and all tests as required by the applicable ASTM Standard. All other welds must be inspected per Section 33 52 23.15 POL SERVICE PIPING WELDING. ASTM A312/A312M seamless piping must be provided with a minimum schedule 10S wall thickness for pipe 200 mm and larger; minimum schedule 40S for pipe smaller than 200 mm (except for threaded pipe which must be minimum Schedule 80S).

- (2) Pipe Ends: All Piping must be provided with beveled ends per Chapter V, ASME B31.3, and must be shipped with the ends capped.
  - (3) Factory Testing and Inspection Records: Per Table K341.3.2 of Chapter IX of ASME B31.3, visual, radiographic and liquid penetrant tests must be performed for each section of piping provided as all sections are subjected to cyclic conditions. All testing and inspections records must be submitted to the Contracting Officer and must indicate the pipe mark and installed location of each piping section on the project site. Observation by the Contracting Officer of the manufacturers and the fields testing and inspection procedures must be allowed under this contract. Pipe certification along with pipe markings must be submitted before the pipe arrives on the job site.
  - (4) Qualifications of Owner's Inspector: Provide a qualified inspector in accordance with Chapter VI of ASME B31.3. to act as the owner's inspector (for the Government) at the pipe manufacturer's facility in addition to the manufacturer's inspector.
  - (5) Quality Assurance Plan: Submit Quality Assurance Plan for the welding, inspecting and testing of the welded seam pipe.
- b. Stainless Steel Control Tubing: Seamless, fully annealed tubing conforming to ASTM A269/A269M, Grade TP316, Rockwell hardness B80 or less. Wall thickness for 13 mm tubing to be a minimum of 1.2 mm.

#### 2.2.4 External Protective Coatings for Aboveground Piping

Provide exterior coating of aboveground piping and fittings, piping in pits, pipe supports, filter separators, and miscellaneous metal and system components in accordance with Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Color of finish coat must be white. Do not coat aboveground stainless steel or aluminum surfaces.

#### 2.2.5 External Protective Coatings for Buried Steel Piping

##### 2.2.5.1 Carbon Steel Piping

- a. New pipe and fittings must be factory coated in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEM.
- b. Field joints and repairs must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- c. Field joints and repairs in tight spots (valve pits when heaters are too big) must be liquid epoxy in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.

- d. Existing systems must match existing coating system and must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- e. Abrasion-resistant topcoat. Following the initial FBE coating application, provide a 20 mil thick abrasion-resistant FBE topcoat. Abrasion-resistant topcoat must be specifically suited for directional boring piping installation.

## 2.2.6 Fittings

### 2.2.6.1 General

Welding ells, caps, tees, reducers, must be of materials compatible for welding to the pipeline in which they are installed, and wall thickness, pressure and temperature ratings of the fittings must be not less than the adjoining pipeline. Unless otherwise specified herein or required by the conditions of installation, all elbows must be the 1.5 diameter (D) type. Miter joints are not acceptable. Make odd angle offsets with pipe bends or elbows cut to the proper angle. Butt weld fittings must be factory-made wrought fittings manufactured by forging or shaping. Fabricated fittings will not be permitted. Welding branch fittings must be insert type suitable for radiographic inspections specified herein.

Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 65 mm or less in diameter, and the branch outlet is located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas equipped with containment curb.

### 2.2.6.2 Carbon Steel Fittings

- a. Fittings 65 mm and Larger: Butt weld, conforming to ASTM A234/A234M, grade WPB and ASME B16.9 of the same wall thickness as the adjoining pipe. All welds must on buried piping be radiographically examined throughout the entire length of each weld. Each fitting must be subjected to the Supplementary Requirements S3 and S4, Liquid Penetration examination and Magnetic-Particle Examination. Detectable flaws will not be accepted in the supplementary examinations. Fittings must be identified to relate them to their respective radiograph.
- b. Fittings 50 mm and Smaller: Forged, butt weld or socket welded (except flanges, see below). If specifically indicated on drawings, non-flange fittings may be threaded. Socket welded or threaded fittings must be Class 3000, conforming to ASTM A105/A105M, Grade 2 and ASME B16.11. Threaded fittings must only be used for above grade applications. Underground and in pits, low point drain pipe, and high point vent pipe must be butt welded.
- c. Flanges: Class 150 or Class 300 (as indicated), weld neck, butt weld, forged flanges conforming to ASTM A105/A105M, and ASME B16.5 except flanges that are to be connected to pumps must match the pump flanges rating. Threaded and slip-on flanges are not allowed. Flanges to be

2 mm raised face with modified spiral serrated gasket surface finish, except where required otherwise to match system components furnished. Match flange face to valves or system components furnished. Flange face must be machined to match valves or system components furnished. Use of spacing rings or gaskets discs are not allowed. Flanges must be subjected to the Supplementary Requirements S56, Liquid Penetrant Examination as outlined in ASTM A961/A961M. Detectable flaws will not be accepted. For flanges 50 mm and smaller located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas the fitting may be forged (socket welded), Class 150 or Class 300, (conforming to ASTM A105/A105M, Grade 2) and ASME B16.5. All flanges must be radiographicable, butt welded, weld neck type.

d. Piggable System (JP-5 Pipeline)

- (1) Provide barred tees on all branch outlets 50 mm and larger when within 6.1 meters of pig launcher or receiver barrel, including the barrels. Provide barred tees on all size outlets greater than 50 mm in size with any part of the outlet on the bottom half of the pipe. Provide barred tees on all branch connections equal to or greater than 50 percent of piggable line size.
- (2) Use 3 D sweeps between pig launchers and receivers.

2.2.6.3 Stainless Steel Fittings

- a. Fittings 65 mm and Larger: Butt weld stainless steel conforming to ASTM A403/A403M, Class WP, Type 304L, seamless or welded, and ASME B16.9 of the same minimum wall thickness as the adjoining pipe. Welded fittings must be tested and inspected the same as the welded seam pipe and meet the same requirements as for the pipe.
- b. Fittings 50 mm and Smaller: Forged Type 304 or 304L, butt weld, or socket welded (except flanges, see below). If specifically indicated on drawings, non-flange fittings may be threaded. Socket welded or threaded fittings must be Class 3000 conforming to ASTM A182/A182M and ASME B16.11. Threaded fittings must only be used for above grade applications. Underground and in pits, low point drain pipe, and high point vent pipe must be butt welded.
- c. Unions: Conforming to ASTM A182/A182M, Grade 304 or 316.
- d. Flanges. Class 150 weld neck, butt weld, forged Type 304 stainless steel flanges conforming to ASTM A182/A182M and ASME B16.5, except flange that are to be connected to pumps must match the pump flanges rating. Threaded and slip-on flanges are not allowed. Flanges to be 2mm raised-face with modified spiral serrated gasket surface finish, except where required otherwise to match system components furnished. Flange face must be machined to match valves or system components furnished. Match flange face to valves or system components furnished. Flanges must be subjected to the Supplementary Requirements S56, Liquid Penetrant Examination as outlined in ASTM A961/A961M. Detectable flaws will not be acceptable. For flanges 50 mm and smaller located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas the fitting may be forged (socket welded), Class 150, conforming to ASTM A182/A182M and ASME B16.5. In pits, vaults, on thermal relief valve piping for pipeline routes, and other uncontained locations the

flanges must be radiographicable, butt welded, weld neck type.

e. Piggable System (JP-5 Pipeline):

(1) Provide barred tees on all branch outlets 50 mm and larger when within 6 m of pig launcher or receiver barrel, including the barrels. Provide barred tees on all size outlets greater than 50 mm in size with any part of the outlet on the bottom half of the pipe. Provide barred tees on all branch connections equal to or greater than 50 percent of piggable line size.

(2) 3 D sweeps between pig launchers and receivers.

f. Stainless Steel Tube Fittings. Flareless, 316 stainless steel fittings conforming to SAE J514.

2.2.7 Insulating Flange Kits (Electrically Isolating)

Provide ASTM D229 electrical insulating material of 1,000 ohms minimum resistance or 500 Volts per mil (VPM) minimum dielectric strength; material must be resistant to the effects of aviation and non-aviation hydrocarbon fuels. Provide full face insulating gaskets between flanges. Provide full surface 0.75 mm thick wall thickness, spiral-wound mylar insulating sleeves between the bolts and the holes in flanges; bolts may have reduced shanks of a diameter not less than the diameter at the root of threads. Provide 3 mm thick high-strength phenolic insulating washers next to flanges and provide flat circular stainless steel washers over insulating washers and under bolt heads and nuts. Provide bolts 12 mm longer than standard length to compensate for the thicker insulating gaskets and the washers under bolt heads and nuts. Above grade flanges separated by electrically insulating flange kits must be provided with weatherproof lightning surge arrester devices. The surge arrester must bolt across flanges separated by insulating gasket kits per detail on contract drawings. Provide with flange protector as described in this section. The arrester must have the following features:

- a. Weatherproof NEMA 4 enclosure.
- b. Bidirectional and bipolar protection.
- c. Constructed of solid state components, no lights, fuses or relays and used without required maintenance or replacement.
- d. Withstand unlimited number of surges at 50,000 Amperes.
- e. Maximum clamping voltage of 700 Volts based on a IEEE C62.41.2 8x20 microsecond wave form at 50,000 Amperes peak measured at the device terminals (zero lead length).
- f. A UL listed arrester for installation in Class 1, Division 1, Group D, hazardous areas.

Install the mounting bracket and leads on the flange side of the bolt insulating sleeve and washer, and size in accordance with this schedule:

| Line Size                                                                                                            | Bolt Size |
|----------------------------------------------------------------------------------------------------------------------|-----------|
| 50 mm                                                                                                                | 16 mm     |
| 65 mm                                                                                                                | 16 mm     |
| 80 mm                                                                                                                | 16 mm     |
| 100 mm                                                                                                               | 16 mm     |
| 150 mm                                                                                                               | 19 mm     |
| 200 mm                                                                                                               | 19 mm     |
| 250 mm                                                                                                               | 22 mm     |
| 300 mm                                                                                                               | 22 mm     |
| 350 mm                                                                                                               | 25 mm     |
| 400 mm                                                                                                               | 25 mm     |
| Note: Make allowance for the 1 mm thickness of the insulating sleeve around the bolts when sizing the mounting lugs. |           |

#### 2.2.8 Bolts, Nuts and Washers

- a. Bolts and nuts for pipe flanges, flanged fittings, valves and accessories must conform to ASME B18.2.1 and ASME B18.2.2, except as otherwise specified.
- b. Bolts must be of sufficient length to obtain full bearing on the nuts and must project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque.
- c. Bolts must be regular hexagonal bolts conforming to ASME B18.2.1 with material conforming to ASTM A193/A193M, Class 2, Grade B8, stainless steel, when connections are made where a stainless steel flange is involved, and Grade B7, chromium molybdenum alloy, when only carbon steel flanges are involved. Bolts and nuts chosen must have sufficient strength to seat gasket types chosen. Bolts must be threaded in accordance with ASME B1.1, Class 2A fit, Coarse Thread Series, for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- d. Nuts must conform to ASME B18.2.2, hexagonal, heavy series with material conforming to ASTM A194/A194M, Grade 8, stainless steel for stainless steel bolts, and Grade 7, chromium molybdenum alloy for chromium molybdenum alloy bolts. Nuts must be threaded in accordance with ASME B1.1, Class 2B fit, Coarse Thread Series for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- e. Provide washers under bolt heads and nuts. Use chromium molybdenum alloy washers dimensioned to ASTM F436 flat circular for chromium molybdenum bolts. Stainless steel washer dimensioned similar to ASTM F436 flat circular, use material the same as the bolt.
- f. Use torque wrenches to tighten all flange bolts to the torque recommended by the gasket manufacturer. Tight in the pattern recommended by the gasket manufacturer. Use anti-seize compound on

stainless steel bolts.

#### 2.2.9 Flange Gaskets, Non-Metallic, Non-Electrically Isolating

ASME B16.21, composition ring, using a Nitrile Rubber such as Buna-N and NBR, polytetrafluoroethylene (PTFE), or a fluoro rubber such as FKM, FPM and Viton®. The gasket must be 3 mm thick. Gaskets must be resistant to the effects of aviation and non-aviation hydrocarbon fuels and manufactured of fire-resistant materials. Full-face gaskets must be used for flat-face flanged joints. Ring gaskets must be used for raised-face flanged joints. Gaskets must be of one piece factory cut. Select a gasket suitable for the working and test pressure of the fluid.

##### 2.2.9.1 Nitrile Butadiene (Buna-N)

Provide Buna-N material that conforms to SAE AMS3275.

##### 2.2.9.2 Acrylonitrile Butadiene Rubber (NBR)

Provide NBR material that conforms to SAE AMS3275.

##### 2.2.9.3 Polytetrafluoroethylene (PTFE)

Provide PTFE material that conforms to ASTM F336.

##### 2.2.9.4 Fluoro Rubber FKM

Provide FKM material that conforms to ASTM D1418.

##### 2.2.9.5 Fluoroelastomer FPM

Provide FPM material that conforms to ISO 1629.

#### 2.3 MANUAL VALVES

All portions of a valve coming in contact with fuel in stainless steel pipelines or epoxy lined carbon steel pipelines must be of noncorrosive material. Valves in stainless steel pipelines or epoxy lined carbon steel pipelines must be Type 304 or Type 316 stainless steel or carbon steel internally plated with chromium or nickel or internally electroless nickel plated. Valves in unlined carbon steel pipelines must have carbon steel body. Stem and trim must be stainless steel for all valves. Manually operated valves 150 mm and larger must be worm-gear operated and valves smaller than 150 mm must be lever operated or handwheel operated. Valves smaller than 50 mm must have lever-type handles. Valves installed more than 2.4 m above finished floor must have chain operators and a position indicators visible from ground level. Sprocket wheel for chain operator must be aluminum. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits must be full bore, piggable, double block and bleed type. The full bore piggable valves at the launcher and the receiver must be ball type. Valves must be true full bore with no projections extending into the flow path of the pig train.

##### 2.3.1 Ball Valves

Ball valves must be fire tested and qualified in accordance with the requirements of API Std 607 and API STD 608. Seal material for the fire test must be graphite, seal material for the project must be as as

indicated below. Ball valves must be nonlubricated valves that operate from fully open to fully closed with 90-degree rotation of the ball. Valves 50 mm and larger must conform to applicable construction and dimension requirements of API Spec 6D, ANSI Class 150 or Class 300 (as indicated) and must have flanged ends. The balls in valves 250 mm and larger full port and 300 mm and larger regular port and larger must have trunnion type support bearings. Except as otherwise specified or indicated, reduced port or full port valves may be provided at the Contractor's option. Balls must be solid, not hollow cavity.

#### 2.3.1.1 Materials

Ball must be stainless steel. Ball valves must have tetrafluoroethylene (TFE) or fluoroelastetomer (FKM), commonly referred to as Viton seats, body seals and stem seals. Valves 100 mm and smaller must have a locking mechanism.

#### 2.3.1.2 Full Port Ball (DBBV) Valves for Piggable Lines (JP-5 Pipeline)

Required for piggable lines, and where indicated elsewhere. Ball valves must be designed, manufactured, and tested to API Spec 6D, fire-safe and tested to API Spec 6FA, and BS EN ISO 10497 (BS 6755, Part 2). Seal material for the fire test must be graphite, seal material for the project must be as as indicated below. Valves must be trunnion-mounted with independent spring and hydraulically actuated, floating, single piston effect, self-relieving seat rings, with bi-directional sealing. Ball must be solid type with full through-conduit opening, suitable for passage of pipeline pigs. Stem must be anti-static, blow-out-proof design with o-ring seals and provided with an emergency sealant injection fitting. Valves must be 3-piece, bolted body design with raised-faced ANSI Class 150 or Class 300 (as indicated) flanged connections, equipped with body drain/bleed valve and vent fitting, and suitable for double block and bleed service in the closed and open positions. Valves must be all stainless steel construction, or carbon steel with stainless steel stem, and all wetted parts electroless nickel-plated. Valves must have nylon, devlon, or polytetrafluoroethylene (TFM) seat inserts, FKM B body, stem, and seat o-rings, with stainless steel and graphite body gaskets and graphite secondary stem seals. Valves located in vaults or pits must be equipped with actuator extensions.

#### 2.3.1.3 Electric Valve Actuator

Electric valve actuator must be as indicated for Plug (Double Block and Bleed) Valves, electric valve actuator.

#### 2.3.2 Plug (Double Block and Bleed) Valves

API Spec 6D, API Spec 6FA, ANSI Class 150 or Class 300 (as indicated), non-lubricated, resilient, double seated, trunnion mounted, tapered lift plug capable of two-way shutoff. Valve must have tapered plug of steel or ductile iron with chrome or nickel plating and plug supported on upper and lower trunnions. Sealing slips must be steel or ductile iron, with Viton seals which are held in place by dovetail connections. Valve design must permit sealing slips to be replaced from the bottom with the valve mounted in the piping. Valves must operate from fully open to fully closed by rotation of the handwheel to lift and turn the plug. Valves must have weatherproof operators with mechanical position indicators. Indicator shaft must be stainless steel. Minimum bore size must be not less than 65 percent of the internal cross sectional area of a pipe of the same nominal

diameter unless bore height of plug equals the nominal pipe diameter and manufacturer can show equal or better flow characteristics of the reduced bore size design. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits, in fuel piping between the pig launchers and the pig receivers must be full bore, piggable. Valves must be true full bore with no projections extending into the flow path of the pig train. Full port plug valves in distribution piping must be provided with a 15 mm threaded body drain.

#### 2.3.2.1 General

Valves in the operating tank suction and fill lines and the valves at the four valve manifold in the pump room in the tank fill lines must be provided with a factory-installed limit switch that is actuated by the valve closure. Tank fill line valve and four valve manifold limit switches must be provided with one double pole double throw contacts or four single pole, double throw contracts, two for open, two for closed. Tank suction line valve limit switches must be provided with one double pole double throw contacts or four single pole, double throw contacts, for closed, and one single pole double throw contact or two single pole, double throw contacts for open. All components must be watertight and U.L. listed for Class I, Division 1, Group D hazardous areas.

#### 2.3.2.2 Valve Operation

Rotation of the handwheel toward open must lift the plug without wiping the seals and retract the sealing slips so that during rotation of the plug clearance is maintained between the sealing slips and the valve body. Rotation of the handwheel toward closed must lower the plug after the sealing slips are aligned with the valve body and force the sealing slips against the valve body for positive closure. When valve is closed, the slips must form a secondary fire-safe metal-to-metal seat on both sides of the resilient seal. Plug valves located in Isolation Valve Pits or vaults must be provided with handwheel extensions.

#### 2.3.2.3 Relief Valves

ANSI Class 150 or Class 300 (as indicated). Provide plug valves with automatic thermal relief valves to relieve the pressure build up in the internal body cavity when the plug valve is closed. Relief valves must open at 175 kPa differential pressure and must discharge to the throat of, and to the upstream side, of the plug valve.

#### 2.3.2.4 Bleed Valves

ANSI Class 150 or Class 300 (as indicated), stainless steel body valve. Provide manually operated bleed valves that can be opened to verify that the plug valves are not leaking when in the closed position.

#### 2.3.2.5 Electric Valve Actuator

The actuator, controls and accessories must be the responsibility of the valve-actuator supplier for sizing, assembly, certification, field-testing and any adjustments necessary to operate the valve as specified. The electric valve actuator must include as an integral unit the electric motor, actuator unit gearing, limit switch gearing, position limit switches, torque switches, drive bushing or stem nut, declutch lever, wiring terminals for power, remote control, indication connections and

handwheel. The electrically actuated plug valve must be set to open and close completely in 30 to 60 seconds against a differential pressure of 2 MPa. The actuator settings of torque and limit contacts must be adjustable. The valve actuator must be suitable for mounting in a vertical or horizontal position and be rated for 30 starts per hour. The valve actuator must be capable of functioning in an ambient environment temperature.

- a. The electrical enclosure must be specifically approved by UL or Factory Mutual for installation in Class I, Division 1, Group D locations.
- b. The electric motor must be specifically designed for valve actuator service and must be totally enclosed, non-ventilated construction. The motor must be capable of complete operation at plus/minus 10 percent of specified voltage. Motor insulation must be a minimum NEMA Class F. The motor must be a removable subassembly to allow for motor or gear ratio changes as dictated by system operational requirements. The motor must be equipped with an embedded thermostat to protect against motor overload and also be equipped with space heaters. It must de-energize when encountering a jammed valve.
- c. The reversing starter, control transformer and local controls must be integral with the valve actuator and suitably housed to prevent breathing or condensation buildup. The electromechanical starter must be suitable for 30 starts per hour. The windings must have short circuit and overload protection. A transformer, if needed, must be provided to supply all internal circuits with 24 VDC or 110 VAC may be used for remote controls.
- d. The actuator gearing must be totally enclosed in an oil-filled or grease-filled gearcase. Standard gear oil or grease must be used to lubricate the gearcase.
- e. The actuator must integrally contain local controls for Open, Close and Stop and a local/remote three position selector switch: Local Control Only, Off, and Remote Control plus Local Stop Only. A metallic handwheel must be provided for emergency operation. The handwheel drive must be mechanically independent of the motor drive. The remote control capability must be to open and close. Rim pull to operate valve manually must not exceed 28 kg.
- f. Position limit switches must be functional regardless of main power failure or manual operation. Four contacts must be provided with each selectable as normally open or normally closed. The contacts must be rated at 5A, 120 VAC, 30 VDC.
- g. Each valve actuator must be connected to a PLC supplied by "others."
- h. The actuator must have a local display of position even when power has been lost.
- i. The actuator must be supplied with a start-up kit comprising installation instruction, electrical wiring diagram and spare cover screws and seals.
- j. The actuator must be performance tested and a test certificate must be supplied at no extra charge. The test should simulate a typical valve load with current, voltage, and speed measured.

### 2.3.3 Swing Check Valves

Swing check valves must conform to **API STD 600**, regular type, ANSI Class 150 with flanged end connections. Discs and seating rings must be renewable without removing the valve from the line. The disc must be guided and controlled to contact the entire seating surface.

## 2.4 THERMAL RELIEF VALVE

### 2.4.1 Valve Material

Valves must have carbon steel bodies (stainless steel on stainless steel pipelines) and bonnets with stainless steel springs and trim. Valves must be Class 150 or **Class 300 (as indicated)** flanged end connections.

### 2.4.2 Thermal Relief Valve (ASME Type)

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, hydraulically operated type with plain caps, and must be labeled in accordance with **ASME BPVC SEC VIII D1**. Valve stems must be fully guided between the closed and fully opened positions. The valves must be factory-set to open at pressures indicated on the drawings. Operating pressure must be adjustable by means of an enclosed adjusting screw. The valves must have a minimum capacity of 20 GPM at 10 percent overpressure. Valves must have a replaceable seat. Relief valves that do not relieve to a zone of atmospheric pressure or tank must be a balanced type relief valve.

### 2.4.3 Thermal Relief Valve (Balanced Type)

Thermal relief valves that do not relieve to a zone of atmospheric pressure or atmospheric tank must be a balanced type relief or regulator valve.

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, fully balanced type (back pressure must not affect relief pressure) back pressure regulator/relief valve. Set valve at pressure indicated on drawings. Valve body must have **25 mm** (minimum) raised face flange connections unless otherwise indicated. Orifice must have a minimum orifice size of **15 mm** in diameter. Valve must have bubble-tight piston and seat design with stainless steel piston and Viton seat. Valve must be selected for the nominal flow condition of: pass a minimum of **18 liters per minute**, at a differential pressure of **758 kilopascal**, with a nominal set pressure of **689 kilopascal**. Valve must be factory configured to open at required set pressure but must be field adjustable by means of an enclosed adjusting screw **for set pressures up to 1034 kilopascals**.

## 2.5 PIPING ACCESSORIES

### 2.5.1 Mechanically Adjustable Segmented Elastomeric Seal

Mechanically adjustable segmented elastomeric seals must be constructed of fuel resistant Buna-N elastomers and Type 316 stainless steel fasteners and hardware.

### 2.5.2 Pipe Sleeves

Pipe sleeves must be installed where indicated and at all points where the piping passes through concrete construction. Such sleeves must be of sufficient inside diameter to provide a minimum clear distance between the pipe and the sleeve of 13 mm. Sleeves through concrete pits or slabs must be standard weight carbon steel pipe with a protective coating. Each sleeve must extend through the respective pit wall or slab and must be provided with a wrap around Buna-N end seal (boot). (Viton when exposed to sunlight) and secured to the pipe sleeve and piping with adjustable stainless steel hose clamps. Sleeves where piping passes under roads or piping indicated to be double walled must be standard weight carbon steel pipe with a protective coating as previously specified. Alignment of the sleeve and piping must be such that the pipe is accurately centered within the sleeve by a nonconductive centering element. The sleeve must be securely anchored to prevent dislocation. Closure of space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush.

### 2.5.3 Strainers

#### 2.5.3.1 Basket Type

Strainer must be single basket type arranged in a simplex configuration as indicated in compliance with MIL-PRF-13789, except as specified otherwise. Strainer end connections must be designed in accordance with ASME B16.5, Class 150 or Class 300 (as indicated). Strainer body material must be the same as the material specified for manual valves. Strainers must have removable baskets of 40 mesh wire screen with larger wire mesh reinforcement; wire must be stainless steel, Type 316. Pressure drop for clean strainer must not exceed 21 kPa at maximum design flow rate. The ratio of net effective strainer area to the area of the connecting pipe must be not less than three to one. Each strainer must be provided with a suitable drain at the bottom, equipped with a ball valve. The strainer must be equipped with a direct-reading, piston type differential pressure gauge that measures the differential pressure across the basket as perin accordance with Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

#### 2.5.3.2 Cone Type

Strainer must be stainless steel type 304 or 316, 100 100 mesh screen with the ratio of net open area of strainer to the area of the connecting pipe must be not less than three to one at the pump suction. Pump suction strainer must have a 100 mesh screen with not less than 300 percent open area (ratio of the strainer open area to the cross section of pipe).

#### 2.5.4 Pressure Gauge

See Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

### 2.5.5 Pipe Supports

#### 2.5.5.1 General

Pipe supports must conform to MSS SP-58. Design pipe supports to meet the applicable requirements of ANSI/ASME B31.3 or ANSI/ASME B31.4. Provide hot-dip galvanized finish on rods, nuts, bolts, washers, and supports. Provide miscellaneous metal that conforms to ASTM A36/A36M, standard mill finished structural steel shapes, hot-dipped galvanized.

#### 2.5.5.2 Adjustable Pipe Supports

Adjustable pipe supports must consist of a cast iron saddle and a threaded nipple connected to a carbon steel pipe by means of a special reducer conforming to MSS SP-58. The supports must be provided with PTFE insulation strips.

#### 2.5.5.3 U-bolt Half Round Supports

Supports must have anti-friction bearing half-round in contact with the bottom of the pipe. Provide Polytetrafluoroethylene or like hydrophobic, anti-corrosive material half-round with a compressive strength of at least 69 MPa or greater as required. U-bolts must be installed in either a loose or limited guide configuration as indicated on the design drawings. Provide hot-dip galvanized u-bolts with seamless non-metallic low friction coating. U-bolt connection must be double nutted on the bottom and single bolted on top.

#### 2.5.5.4 Concrete and Grout

Concrete and grout for anchors and supports must comply with Section 03 30 00 CAST-IN-PLACE CONCRETE.

#### 2.5.6 Sample Connections

- a. Sample connections must be factory assembled units specifically designed for obtaining representative samples from fuel pipelines. Each connection must include a 6 mm sampling probe where the probe faces upstream, ball valve and 6 mm quick disconnect coupling with dust plug, all assembled into a unit that is suitable for installation in a pipe nipple. The sampling probe must extend not less than 25 mm into the fuel pipe. All materials in the sample connections must be stainless steel or aluminum.
- b. Furnish two sampling hose assemblies to the Contracting Officer at the project site. Each assembly must consist of a 1.8 m length of 6 mm clear plastic tubing with internal bonding/grounding wire. One end of the tubing will contain a male connector that actuates flow when inserted into the quick disconnect coupler. Each end of the bonding/grounding wire must be equipped with clips for attaching to the pipe and metal sample container.

#### 2.5.7 Sight Flow Indicators

Sight flow indicators must be ANSI Class 150 and must have flanged end connections. Sight flow indicators must consist of a housing containing a rotating propeller that is visible through a glass observation port. The housing must be stainless steel when installed in stainless steel lines and carbon steel when installed in carbon steel lines. The glass in the indicator must also meet the Class 150 rating.

#### 2.5.8 Volumetric Turbine Flow Meter

Volumetric turbine flow meter must be a turbine type meter designed for a continuous flow of 1200 to 12,000 BPH, constructed of 316 stainless steel. The turbine meter must be supplied with Class 300 stainless steel ASME flanges, be capable of 2520 kPa system pressure. Meter must be factory calibrated for JP-5 fuel. The measuring element of the turbine

will consist of a straight blade, un-rimmed central rotor, rotating about a central rotor shaft that is supported bilaterally within the inside diameter of the meter body by cylindrical shaped spring clips that maintain the shaft and turbine rotor on the center line of the meter body independent of system pressures and temperatures. The cylindrical shaft clips also will counteract swirl and present a uniform fully turbulent flow profile and uniform boundary layer to the cones and turbine rotor. The turbine meter will have an accuracy of plus/minus 0.5 percent over a 10:1 range and a linearity of up to plus/minus 0.25 percent may be attained with premium calibration. Repeatability of the turbine meter will be 0.1 percent of reading over the entire range of the size of the turbine selected. The turbine meter must be approved by US NIST for solvent, gasoline, diesel fuel, fuel oil, and ethanol for use on custody transfer applications. The turbine meter will be supplied complete with an integrally mounted Multi-Function Microprocessor Based Rate Indicator / Totalizer with Field Programmability and Backlit Display in an Aluminum Enclosure. The enclosure must be rated for explosion proof environments. Input power must be self-contained battery, 10-14 VDC or 20-28 VDC. Outputs must include 4-20mA output and pulse output. A reset magnet, aluminum union, and 2-wire Molex signal cable will also be included with the Indicator/Totalizer. The Indicator/Totalizer must have temperature compensation with a four wire RTD input as well as the ability for RS232 data logging.

#### 2.5.9 Differential Pressure Transmitter

Differential pressure transmitter must consist of a capacitance sensor operating on a differential in pressure of fuel. The output must be a 4 - 20mA dc, square root signal between a minimum of 4 - 100 percent of the input. It may be linear between 0 - 4 percent. It simultaneously will produce a digital HART (Highway Addressable Remote Transducer) output signal.

- a. Transmitter body must be stainless steel with stainless steel diaphragm capsule process connecting to a 13 mm 1/2-inch NPT. Drain and vent valves to be stainless steel. Accuracy must be plus/minus 0.20 percent of calibrated span including combined effects of linearity, hysteresis and repeatability.
- b. Display at the transmitter must be LCD. The digital scale must be a 4-digit LCD, capable of being read in low light/no light conditions. Indicator scale must be in L per minute.
- c. Differential pressure transmitters must be UL, FM, or CSA listed for Class 1, Division 1, Group D hazardous environment as defined by NFPA 70. Complete installation must be in accordance with manufacturer's recommendations.

#### 2.6 PIGGING SYSTEM COMPONENTS (JP-5 Pipeline System Only)

##### 2.6.1 Smart Pig Launchers and Receivers

Construct of the same materials as the pipe, valves and fittings for a Class 300 system. The length of the straight barrel and the line size section must each be 4.5 meter. Provide associated launcher kicker piping and receipt bypass piping, not less than 250 mm for 400 mm and larger pipelines.

## 2.6.2 Launcher and Receiver Closure Door

The closure must be hinged, swing bolted closure of the same material as the pipe and for a Class 300 system. Gasket must be nitrile or viton. Eye bolts must be pinned to lugs on the hub.

## 2.6.3 Signaler

The pig signaler must be mechanical flag type with manual reset, and be located on the pig launcher and the pig receiver. Material in contact with the fuel must be stainless steel. Units must be suitable for removal and installation under line pressure of 3.0 MPa (440 psi). Signaler must be capable of withstanding line pressure of a Class 300 system.

## 2.7 MISCELLANEOUS ACCESSORIES

### 2.7.1 Concrete Anchor Bolts

Concrete anchors must conform to ASTM A307, Grade C, hot-dipped galvanized.

### 2.7.2 Coatings for Bolts, Studs, Nuts, and Washers

Carbon steel bolts, studs, nuts, and washers must be provided with a factory applied cadmium coating that conforms to ASTM B696 or ASTM B766.

### 2.7.3 Polytetrafluoroethylene (PTFE) Tape

Tape must conform to ASTM D3308.

### 2.7.4 Pipe Sleeves

Provided sleeves constructed of Coated Carbon steel piping, ductile iron, or cast-iron pipe conforming to ASTM A53/A53M, Standard weight.

### 2.7.5 Buried Utility Tape

Provide detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape in minimum 75 mm width rolls, color coded for the utility involved, with warning identification imprinted in bold black letters continuously and repeatedly over entire tape length. Warning identification must be at least 25 mm high and must state as a minimum "BURIED JET FUEL PIPING BELOW". Provide permanent code and letter coloring that is unaffected by moisture and other substances contained in trench backfill material.

### 2.7.6 Pipeline Markers

Provide pipeline markers constructed of 150 mm diameter, one-half inch thick bronze disk with a 75 mm long bronze headed bolt welded to the back of the disk. Engrave the front of the disk with the words "UNDERGROUND FUEL LINE" in the case of one line and "UNDERGROUND FUEL LINES" in the case of multiple fuel lines.

## 2.8 FINISHES

### 2.8.1 Factory Coating

#### 2.8.1.1 Valves

Valve surfaces must be blasted clean according to [SSPC SP 5/NACE No. 1](#). Valve surfaces must be primed and coated in accordance with Section [09 97 13.27](#) HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

#### 2.8.1.2 Equipment and Components

Unless otherwise specified, provide equipment and components fabricated from ferrous metal with the manufacturer's standard factory finish. Each factory finish must withstand 500 hours exposure to the salt spray test specified in [ASTM B117](#). For test acceptance, the test specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond [3 mm](#) on either side of the scratch mark immediately after completion of the test. For equipment and component surfaces subject to temperatures above [50 degrees C](#), the factory coating must be appropriately designed for the temperature service.

### 2.8.2 Field Painting

Painting required for surfaces not otherwise specified must be field painted as specified in Section [09 97 13.27](#) HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Do not paint aboveground stainless steel and aluminum surfaces. Do not coat equipment or components provided with a complete factory coating. Hot dip galvanized pipe support steel must not receive the Zinc-Rich Epoxy Primer Coat only the Epoxy Intermediate Coat and Polyurethane Topcoat must be applied to these surfaces. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

## PART 3 EXECUTION

### 3.1 GENERAL

Installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with [ASME B31.3](#) and [NFPA 30](#), except as modified herein. Safety rules as specified in [NFPA 30](#) and [NFPA 407](#) must be strictly observed. Never direct bury threaded connections, socket welded connections, unions, flanges, valves, air vents, or drains. Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.

### 3.2 VERIFICATION OF DIMENSIONS

After becoming familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

### 3.3 CLEANING OF PIPING

Keep the interior and ends of all new piping, affected by construction operations, thoroughly cleaned of foreign matter and water before and after being installed. Piping systems must be kept clean during installation by means of plugs or other approved methods. When work is not in progress, open ends of piping and fittings must be closed so that no water or other foreign substance will enter the pipes or fittings.

Piping must be inspected before placing into position. The interior of each length of pipe must be cleaned after welding ensuring that the interior of the piping is free of foreign matter when it is connected into the system.

### 3.4 TRENCHING AND BACKFILLING

Trenching and backfilling must conform to Section 31 00 00 EARTHWORK, and the following bedding and backfill requirements. The pipe must be laid in a bed of sand 150 mm deep, compacted in accordance with Section 31 00 00 EARTHWORK, paragraph BACKFILLING AND COMPACTION. Sand must meet the requirements of Section 31 00 00 EARTHWORK, paragraph SELECT GRANULAR MATERIAL. The full length of each section of pipe without any protective covering must be excavated to permit installation of the protective covering. Pipe that has the grade or joint disturbed after laying, must be taken up and relaid. Pipe must not be laid in water or when the trench or weather conditions are unsuitable for such work. After testing and application of protective covering to joints, sand backfill must be placed and compacted around the pipe or protective coating to a depth of 300 mm above top of pipe. The remainder of the backfill must be the same as for other types of pipe.

### 3.5 PIPING LAYOUT REQUIREMENTS

#### 3.5.1 Pipe Fabrication

Fabricate piping to measurements established on the project site and position into place without springing or forcing. Make provisions for absorbing expansion and contraction without undue stress in any part of the system. The use of flexible connectors in permanently mounted pump suction and discharge lines as a method of compensating for piping misalignment is not acceptable.

#### 3.5.2 Interferences and Measurements

Provide offsets, fittings, and accessories required to eliminate interferences and to match actual system components connection locations and arrangements. Verify measurements before commencing work. Submit discrepancies for clarification before proceeding with the installations to the Contracting Officer.

#### 3.5.3 Space and Access

Keep piping, control tubing, which is not detailed close to structures and columns so as to take up a minimum amount of space. Ensure that access is provided for maintenance of system components, valves and gauges.

#### 3.5.4 Location

Do not place unions in locations that will be inaccessible after the completion of the work. Place unions on each side of equipment.

#### 3.5.5 Pipe Supports

Where provided hot dip galvanized pipe support steel must be surface prepped to SSPC SP 16, Brush-Off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals. Once the steel is prepped, apply only the Epoxy Intermediate Coat and Polyurethane Topcoat per manufacturer's instructions.

Provide pipe supports with the maximum spacing as defined in Table 1 below, unless otherwise indicated. At the indicated locations. Provide additional pipe supports at concentrated piping loads (valves).

| Table 1. Maximum Support Spacing |       |        |       |       |        |        |        |        |        |
|----------------------------------|-------|--------|-------|-------|--------|--------|--------|--------|--------|
| Nominal Pipe Size (mm)           | 25 mm | 40 mm  | 50 mm | 80 mm | 100 mm | 150 mm | 200 mm | 250 mm | 300 mm |
| Maximum Support Spacing (m)      | 2 m   | 2.75 m | 3 m   | 3.5 m | 4.25 m | 5 m    | 5.75 m | 6.50 m | 7 m    |

Provide anchors where required to absorb or transmit thrust or eliminate vibration or pulsation. Provide supports near each change of direction. Select support components which do not restrict the movement of the pipe due to thermal expansion. Space supports uniformly and arrange symmetrically.

### 3.5.6 Structural Support

Provide supplementary or intermediate steel or other structural members as required for transmission of loads to members forming part of the supporting structure. Piping must not be supported from other piping.

### 3.5.7 Grade

Where profiles of piping lines are shown on the drawings, grade the line uniformly between changes in slope or direction. Maintain gradient to within 6 mm over the entire length of pipe. When backfilling has been completed to the top of the pipe, the pipe must be surveyed at each joint, and logged by station number. Submit to the Contracting Officer for approval the survey final elevations before backfilling can continue.

### 3.5.8 Size Changes

Make changes in pipe size with reducing fittings. Do not use bushings. Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 65 mm or less in diameter, and the branch outlet is located either in a pump house or on a system component pad equipped with containment curb.

### 3.5.9 Direction Changes

Make changes in direction of pipes with 1.5 D fittings. Where piping is to be piggable, make changes in direction with 3 D sweeps as indicated. Provide special fittings when required. Make odd-angle offsets with pipe bends or elbows cut to the proper angle. Make changes in direction with

fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees will not be permitted. The centerline radius of bends must not be less than 6 diameters of the pipe if a pipe bender is used.

#### 3.5.10 Threaded End Connections

Provide threaded end connections only on piping 50 mm in nominal size or smaller and only where indicated on the drawings. Provide threaded connections with PTFE tape or equivalent thread-joint compound applied to the male threads only. Not more than three threads must show after the joint is tightened.

#### 3.5.11 Existing Pipe Systems

No interruptions or isolation of an existing fuel handling service or system may be performed unless the actions are appropriately documented in the approved work plan. Perform initial cutting of existing fuel pipe with a multiwheel pipe cutter, using a nonflammable lubricant. After cut is made, seal interior of piping with a gas barrier plug. Refer to API RP 2009 and API RP 2200. Purge interior of piping with carbon dioxide or nitrogen prior to performing any welding process.

#### 3.5.12 Bolted Connections

For each bolted connection of stainless steel components (e.g., pipes, piping components, valves, and system components) use stainless steel bolts or studs, nuts, and washers. For each bolted connection of carbon steel components, use carbon steel bolts or studs, nuts, and washers. Bolts project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque. Prior to installing nuts, apply a compatible anti-seize compound to the male threads.

#### 3.5.13 Flanges and Unions

Provide flanged joints in each line immediately preceding the connection to system components or material requiring maintenance such as pumps, general valves, control valves, strainers, and other similar items and as indicated. Assemble flanged joints square and tight with matched flanges, gaskets, and bolts.

#### 3.5.14 Manual Valves

Install isolation plug or ball valves on each side of each system equipment, at the midpoint of looped mains, and at any other points indicated or required for draining, isolating, or sectionalizing purpose. Install valves with stems vertically up unless otherwise indicated. Provide individual supports and anchors for each valve.

#### 3.5.15 Air Vents

Provide 40 mm air vents at all high points and where indicated to ensure adequate venting of the piping system.

#### 3.5.16 Drains

Provide 50 mm drains at all low points and where indicated to ensure

complete drainage of the piping. Drains must be schedule 120.

### 3.5.17 Pipe Sleeves

Provide a pipe sleeve around any pipe that penetrates a wall, floor, or slab. Do not install sleeves in structural members except where indicated or approved. Install pipe sleeves in masonry structures at the time of the masonry construction. Sleeves must be of such size as to provide a minimum of 12 mm all-around clearance between bare pipe and the sleeve. Align sleeve and piping such that the pipe is accurately centered within the sleeve by a nonconductive centering element. Securely anchor the sleeve to prevent dislocation. Closure of the space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush. For wall or floor penetrations, extend each sleeve through its respective wall or floor and cut flush with each surface. Seal around sleeves that penetrate through valve or fuel related pits with a Buna-N casing seal. Seal around sleeves that penetrate through non-fire-rated walls and floors in accordance with Section 07 92 00 JOINT SEALANTS. Seal around sleeves that penetrate through fire-rated walls and floors as specified in Section 07 84 00 FIRESTOPPING.

### 3.6 SEISMIC REQUIREMENTS

Support and brace piping and attach valves to resist seismic loads as shown on the drawings. Structural steel required for reinforcement to properly support piping, headers, and system components but not shown must be provided under this section. Material used for support must be as specified under Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

### 3.7 STRUCTURAL ATTACHMENTS

Provide attachment to building structure concrete and masonry by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Apply inserts and anchors with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Structural steel brackets required to support piping, headers, and system components, but not shown, must be provided under this section. Material used for support must be as specified under Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

### 3.8 WELDING

#### 3.8.1 General

All joints, unless indicated otherwise, in carbon steel and stainless steel piping systems must be welded. Do not weld carbon steel to stainless steel. Welding of fuel pipe joints must comply with Section 33 52 23.15 POL SERVICE PIPING WELDING.

### 3.9 INSTALLATION

#### 3.9.1 Precautions

Take special care to ensure that the protective coating on buried pipe is not damaged during installation and that the completed system is free of rocks, sand, dirt, water, weld slag, and foreign objects including construction debris. Take the following steps to ensure these conditions.

- a. Coated pipe must be handled only with canvas or nylon slings or padded clamps. Any coating damaged by improper handling or storage must be repaired as specified.
- b. Pipe brought to the site must be stored on blocks or horses at least 450 mm above the ground and adequately supported to prevent sagging. Padded blocks or horses must be used for coated pipe. The method and height of storing coated pipe must be in accordance with the coating manufacturer's instructions. Pipe ends must be protected and capped against weather at all times, except to accommodate immediate installation.
- c. Visual inspection must be made of the inside of each length of pipe to ensure that it is clear and clean prior to installation.
- d. The open ends of the pipe system must be closed at the end of each day's work or when work is not in progress by use of expansion plugs and must not be opened until the work is resumed.
- e. A swab, with a leather or canvas belt disc to fit the inside diameter of pipe, must be pulled through each length of pipe after welding in place.
- f. Obstruction remaining in the pipe after completion of the system must be removed at the expense of the Contractor.
- g. Plasma cutters and torches are not to be used to make penetrations in the pipe or to cut pipe.
- h. After installation and backfill of the piggable piping is complete and before fuel is put in the pipe, the pipe will be cleaned per the paragraphs CLEANING PIG RUN and WIRE BRUSH RUN in this Section.

### 3.9.2 Protective Coatings for Buried Piping Including Stainless Steel Piping

#### 3.9.2.1 Application of Coating System

Application of coating system must be in accordance with 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.

#### 3.9.2.2 Inspection and Testing

The condition of factory field coated piping must be the responsibility of the Contractor and all damage to the protective covering during transit and handling must be repaired at no additional cost to the Government. All field coating must be subject to approval by the Contracting Officer. The entire pipe must be inspected as specified in sub-paragraph TESTING OF PROTECTIVE COATINGS under paragraph PROTECTIVE COATINGS FOR BURIED STEEL PIPING. The inspection for holidays must be performed just prior to lowering the pipe into the ditch and every precaution must be taken during lowering and backfilling to prevent damage to the protective covering.

#### 3.9.2.3 Damage Repair

Damaged areas of coating must be repaired as specified in the preceding paragraph for fittings.

### 3.10 INTERIOR EPOXY COATING

When internally epoxy lined pipe is cut, the lining must be ground back from the end a minimum of 25 mm but not more than 38 mm.

### 3.11 INSTALLATION OF UNDERGROUND PIPE

Underground fuel pipelines must be pitched as shown on the drawings. Where not indicated they must be pitched a minimum of 50 mm per 30.5 m. 50 mm pipe size valved drain connections must be provided at all low points and 38 mm pipe size valved outlet vent connections must be provided at all high points. Vent and drain lines must terminate in male cam-type locking end connectors with matching female dust covers and installed in pits. The pipe must have cover as shown on the drawings. Drain lines must be installed at the slopes indicated.

#### 3.11.1 Pipe Assembly

Pipe must be strung parallel and adjacent to or above a trench. The pipe must be supported on padded skids during welding and inspection of joints. Protective coating must be inspected and repaired prior to lowering the pipe into the trench. The pipe must be lowered using only canvas or nylon slings. The sling must be dug from underneath the pipe after placements and must not be pulled from underneath the pipe while in contact with it. Care must be taken to prevent damage to the pipe, welded joints or coating and any such damage must be repaired as directed by the Contracting Officer. Pressure testing of the pipe must be done after it has been placed in final position in the trench.

#### 3.11.2 Warning Tapes in Earth Trenches

For the purpose of early warning and identification of buried pipes outside of building walls during future trenching or other excavation, continuous identification tapes must be provided in the trench. Provide metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured for the purpose of early warning and identification of utilities buried below the tape. Tape must be at least 80 mm in width. Color of tape must be as standard with the manufacturer with respect to the type of utility buried below the tape. Tape must have lettering at least 25 mm high with warning and identification imprinted in bold black letters continuously over the entire tape length with not less than the following identification on the tape: BURIED JET FUEL PIPING BELOW. Tape must be installed in accordance with the printed recommendations of the tape manufacturer, as modified herein. Tapes must be buried at a depth of 150 mm from the top of the subgrade or 300 mm below the top surface of earth. Provide permanent color and printing, unaffected by moisture or soil.

#### 3.11.3 Clearances

Install pipe to be clear of contact with other pipes, pipe sleeves, casings, reinforcing steel, conduits, cables, or other metallic structures. Where pipes cross other pipes or structures with a separation of less than 150 mm, install an insulating separator. Protect the pipe from contact with a 300 mm square by 25 mm thick bituminous-impregnated canefiber board.

#### 3.11.4 Protective Coating

When the protective coating on pipe is damaged, the Contracting Officer must be notified and must inspect the pipe before the coating is patched. If the damage to the pipe is deeper than 1.2 mm, the damage must be repaired by welding in accordance with paragraph WELDING. If the pipe is dented, out of round or damaged to the point that welding will not make it good as new, the length of pipe must be rejected.

#### 3.11.5 Pipeline Markers

Provide above underground fuel piping spaced every 90 meters, at tees, and at changes in direction. For sections of underground piping less than 90 meters long, place at midpoint. Provide directly above pipe for single lines and between pipes where pipes run in pairs. Provide additional marker over each mid-line fitting connections for steel reinforced flexible pipe. Cast marker into 450 mm diameter, 300 mm thick concrete plug unless it is set in an area with concrete paving in which case it must be cast into the concrete paving.

### 3.12 SYSTEM COMMISSIONING

For F-24 storage tank system, conform to Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). For JP-5 pipeline system,, conform to Section 33 08 55.02.

### 3.13 TESTING (Both F-24 and JP-5 Systems)

Piping must be tested by pneumatic and hydrostatic pressure. Testing must comply with applicable requirements of ASME B31.3, NFPA 30 and the requirements specified herein. Hydrostatic testing must be performed using fuel as the liquid. Pressure and hydrostatic testing must be performed only after welding inspection has been completed. Labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein must be furnished by the Contractor. Provide provisions to prevent displacement of the piping during testing. Keep personnel clear of the piping during pneumatic testing. Only authorized personnel must be permitted in the area during pneumatic and hydrostatic testing. Isolate system components such as pumps, tanks, filter separators, and meters from the piping system during the testing. Do not exceed the pressure rating of any component in the piping system during the testing. Following satisfactory completion of each test, relieve the test pressure and seal the pipe immediately. Piping to be installed underground must not receive field applied exterior coatings at the joints or be covered by backfill until the piping has passed the final pneumatic tests described herein.

#### 3.13.1 Before Backfilling

Fuel piping must not be backfilled without completing the coating, pneumatic, weld radiograph and other piping examinations and approvals. Fuel piping in trench must not have exposed welds for longer than 30 days.

##### 3.13.1.1 Exterior Coating Holiday Test

Following installation, test the exterior coating of direct buried piping for holidays using high-voltage spark testing in accordance with NACE SP0188. Repair holidays and retest to confirm holiday-free coating. Text must include all existing underground piping exposed for this project.

### 3.13.2 Pneumatic Test

Piping to be installed underground must not receive field applied protective covering at the joints or be covered by backfill until the piping has passed the pneumatic test described herein. To facilitate the tests, isolate various sections of the piping system and test each one separately. Where such sections terminate at flanged valve points, the line must be closed by means of blind flanges in lieu of relying on the valve. Furnish tapped flanges that can be attached to the end of the section of line being tested, and that will permit a direct connection between the piping and the air compressor or pressurizing pump. No taps in the permanent line will be permitted. Furnish all necessary equipment for testing; all gauges must be subject to testing and approval of the Contracting Officer. The air used for pneumatic testing must have a dew point of no more than 5 degrees C. Provide dehumidifying equipment on the suction or discharge side of the air compressor used to provide air for testing. Pressurizing pump must not exceed 4.7 L/s.

#### 3.13.2.1 Pneumatic Test Procedure

Special safety measures, including the wearing of face mask, must be taken during testing under pressure. Only authorized personnel must be permitted in the area during testing. The pneumatic test pressure must be applied in increments. A preliminary 100 kPa (15 psi) test must be applied. Maintain the pressure while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations). If leaks are discovered, repair the leaks accordingly and retest. Repeat process until system is leak-free at 100 kPa (15 psi). The full test pressure must then be applied. Unless otherwise directed by the Contracting Officer, all piping must be tested at a pressure of 175 kPa (25 psi) for not less than 2-hours, during which time there must be no drop in pressure, only pressure rises with temperature. The pressure source must be disconnected during the final test period. Any leaks revealed by the test must be repaired and the test repeated.

### 3.13.3 Hydrostatic Test

Upon completion of pneumatic testing and after backfilling, hydrostatically test each underground piping system with fuel at 2.93 MPa (425 psi) for F-24 piping system; 3.8 MPa (550 psi) for JP-5 pipeline in accordance with ASME B31.3 and API RP 1110, with no leakage or reduction in gauge pressure for four hours. Upon completion of pneumatic testing, hydrostatically test each aboveground piping system with fuel at 1.97 MPa (285 psi) for F-24 piping system; 3.8 MPa (550 psi) for JP-5 pipeline in accordance with ASME B31.3 and API RP 1110, with no leakage or reduction in gauge pressure for four hours. Furnish electricity, instruments, connecting devices, and personnel for test. If fuel is used for testing, comply with all the requirements in Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Upon completion of hydrostatic testing, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days. Defects in work must be corrected at the Contractor's expense, and the test repeated until the work is proven to be in compliance with the Contract requirements.

#### 3.13.4 Soak Testing

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per [API RP 1595](#). Duration of the test must be a minimum of 4 days and maximum of 7 days.

#### 3.13.5 Performance Testing

After the fuel system testing is completed (including pneumatic and hydrostatic testing) the fuel system must be flushed, cleaned and performance tested as specified in Section [33 08 55.01](#) FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). All control valves, both manual and automatic, must be checked for leaks (any area wetted with fuel) and proper operation and adjusted, repaired or replaced to correct any defects.

### 3.14 PIPE PIGGING - CLEANING ([JP-5 Pipeline](#))

#### 3.14.1 General

Track all pigs, using transmitter and receivers, at no less than [805 m](#) increments, but no less than at four locations. The Contractor must prepare a contingency plan for retrieving a stuck pig and repairing any piping deformations. After pigging, plug valves must be flushed of all debris using the drain port at the bottom of the valve.

#### 3.14.2 Use of Fuel in Cleaning Pigging

Cleaning pigging with fuel will take place after the initial receipt of fuel as per Section [33 08 55.01](#) FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Ensure that the fuel that is returned to the storage tanks during the pig runs is free of gross contamination and passes the color assessment method, and meets the requirements of [MIL-STD-3004-1](#). Provide temporary storage tanks for the high particulate and dark color fuel that accumulates in front of and behind each pig. The contractor is responsible for cleaning the off-spec fuel in order to meet the requirements of [MIL-STD-3004-1](#) or dispose of the off-spec fuel off-base.

#### 3.14.3 Cleaning Pig Run

Initially, a proving pig run (foam density [32 kg/m<sup>3</sup>](#) should be performed to ensure the system is fully piggable. Upon completion of the successful proving pig run, the piping system must be cleaned with a standard cleaning pig. This will provide thorough cleaning of the interior of the piping system. Cleaning pig must be the bi-directional disk scraper style with steel body and replaceable polyurethane guiding and sealing disks, as well as gauge plates of 80 percent pipe diameter with [3 mm](#) segmented aluminum fins. The pig body should include bypass nozzles and transmitter cavity. Propellant must be pressurized fuel using portable pumps. The pig must be examined after the initial run for signs of possible pipe damage, interior slag or other adhered particles. Additional runs must be performed until the amount of collected sludge or debris is minimized, as determined by the System Supplier.

#### 3.14.4 Wire Brush Pig Run

After the cleaning pig runs, the piping system must be cleaned with a wire brush style pig. This will remove weld slag and adhered particles from the system. Wire brush pig must be the bi-directional disk style or directional cup style with two circular steel wire brushes. The pig body

should include bypass nozzles and transmitter cavity. Perform wire brush pig runs until the amount of collected weld slag or debris is minimized, as determined by the System Supplier.

### 3.15 PIPE PIGGING VERIFICATION (JP-5 Pipeline)

#### 3.15.1 Geometry/Ultrasonic Tool Reports

After the system is installed and prior to performance testing, a field/preliminary report must be issued and a debrief given to Government personnel onsite on the condition of the piping system that was pigged. This must be comprised of raw data in the form of a PC download or equivalent which shows a continuous scan of each data unit output. Results of a preliminary interpretation of the data must be reported. These must include as a minimum all critical anomalies. A final report must include a description of the principle of operation, explanation of raw data, presentation of raw data, data to be clearly marked with distance traveled scale with classified anomaly location and all identifiable pipeline features, and all anomalies to be classified with locations in summary tabular form, pipe wall thickness survey, as well as the software necessary to read the data. Submittal must be in the form of digital media copied to a CD or DVD (flash drives are unacceptable).

#### 3.15.2 Pipeline Internal Inspection Operations

##### 3.15.2.1 General

The following pigs will be propelled through the pipeline with fuel in order to inspect the pipeline: 1.7 kg density foam swab, combination poly scraper-magnetic, stainless steel wire brush (plastic brush for internally lined piping), aluminum plate gauge, and geometry/ultrasonic tool. Tracking devices must be used on all pigs. At a minimum, the sequence of pig runs must be as follows: 1) foam swab for proving and cleaning, 2) wire brush for cleaning, 3) scraper-magnetic for cleaning, 4) aluminum plate gauge for gauging internal anomalies, 5) scraper-magnetic for cleaning, 6) wire brush for cleaning, 7) scraper-magnetic for cleaning, 8) foam swab for cleaning, (Note: the number of pig flights of each type of cleaning pigs must be determined by the amount and type of debris removed. The conclusion of the cleaning process must be when debris recovered is only that from the pigs themselves. This determination will be determined by the project's System Supplier and the Contracting Officer), 9) geometry/ultrasonic tool. The pipe wall must be continuously monitored on a real-time basis during the geometry/ultrasonic pig run. Anomalies such as patches, couplings, or flanges must also be identified, and the wall thickness given. The geometry/ultrasonic pig's technician will determine if additional runs are necessary. A permanent data set of internal inspection survey findings must be generated.

##### 3.15.2.2 Preparatory Work

The Government will bring to the attention of the Contractor all statutes, rules and regulations relevant to the performance of the work on the site (on Government property) and will also provide the Contractor with a copy of its own site regulations (if any). Provide the pigging vendors with all-available pipeline records and drawings.

##### 3.15.2.3 Pig Load And Launch

The pig must be loaded into the pig launcher by the Contractor. The

method of loading and lodging the front pig cup into the launcher must not involve the use of uncontrolled mechanical force applied to the rear of the pig.

#### 3.15.2.4 Pipeline Operation During Pigging

All pig runs must be made with the line packed with product. The system pumps will be used to propel the pig. The new pig traps will be used for pig launch and retrieval.

#### 3.15.2.5 Brush and Gauging Survey

Run a brush pig at least as often as previously indicated. The brush pig must be designed and provided by the geometry/ultrasonic pig vendor. Additional runs may be required based upon the amount of debris found in the pipeline. The onsite geometry/ultrasonic pig vendor's personnel must determine if additional runs are required. Immediately following the brush pig run and immediately prior to the geometry/ultrasonic survey, run, as a minimum, a single batching pig fitted with a gauge plate equal to 90 percent of the pipeline normal inside diameter. The plate is to be a segmented aluminum disk of 3 mm thickness. The plate gauge pig must also include a tracker and tracking equipment. Track the pig assembly above ground during the operation.

#### 3.15.2.6 Geometry/Ultrasonic Survey

After a satisfactory gauging pig run, the pipeline geometric defects must be determined by a geometry/ultrasonic tool. The geometry/ultrasonic tool must provide accuracy geometric anomaly detection, and bend radius measuring capability. The data obtained must be presented in a PC software format to allow user friendly analysis and presentation. The geometry/ultrasonic tool assembly must be capable of:

- a. Operating in hydrocarbon liquid environment, specifically jet fuel, at a pressure of up to ANSI 300 rating.
- b. Traversing the pipeline with nominal wall thickness and possible bore restrictions down to 90 percent of nominal pipe inside diameter.
- c. Traversing the pipeline length at a speed of between 60 and 100 m/min when propelled by pumped jet fuel. Pressure differential across pig not to exceed 34 kPa.
- d. Traversing through smooth pipe bends as small as 3D (3 pipe diameters) radius and single miter bends of up to 10 degrees change of direction.
- e. Include a tracker and tracking equipment. Track the pig assembly above ground during the operation. The battery life of the tracker must not be less than 72-hours.
- f. Manual loading into the new horizontal pig trap.

The geometry/ultrasonic tool assembly instrumentation performance must be capable of:

- a. Battery life to be minimum 18-hours at operating conditions.
- b. Principle of operation to be electronically stored geometry system.

- c. Geometry sensing to span full circumference and length of pipe, with associated distance measuring method.
- d. Geometry system must be capable of:
  - (1) positive location and identification of each geometric anomaly.
  - (2) positive location and identification of each bend.
  - (3) positive location and identification of distance marker reference points of either magnetic or electronic type placed on or above the pipe.
- e. Classification of geometric anomalies to be as minimum:
  - (1) discrimination between ovality and intrusion anomalies.
  - (2) mechanical damage such as mill defects, dents, internal gouges, and buckles.
  - (3) pipeline weld defects (such as excess weld penetration).
  - (4) geometric thickness anomalies. As a minimum, these must be reported in the following categories within the listed accuracy.
    - (a) magnitude of anomaly (plus/minus 25 mm)
    - (b) span of anomaly (plus/minus 25 mm)
    - (c) ovality (plus/minus 2.5 mm)
    - (d) span of ovality (plus/minus 25 mm)
    - (e) anomaly station (plus/minus 1:2,000)

#### 3.15.2.7 Pipe Wall Thickness Survey

The geometry/ultrasonic tool must provide accuracy measurement of pipe wall thickness (plus/minus 0.25 mm). The data obtained must be presented in a PC software format to allow user friendly analysis and presentation.

#### 3.15.2.8 Lost Pig

The Contractor is responsible for a lost pig, finding the pig, retrieval of the pig, and all repairs, radiographs to the pipeline system and the pig.

### 3.16 DEMONSTRATIONS

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

### 3.17 POSTED OPERATING INSTRUCTIONS

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

-- End of Section --